Preprint 1–61, 2026

A Loose Ceiling, Not a Certificate: Auditing Fisher-Separability Bounds on Hallucination Detectability Across Architectures

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Abstract

Can the ceiling on hallucination detectability be read off a representation before any probe is trained? Fisher separability J yields a closed-form, training-free value that empirically upper-bounds linear-probe AUROC via the binormal identity $\Phi(\sqrt{J/2})$. Rerunning six models (GPT-2 117M/Medium, Qwen2.5 0.5B–7B) under one matched real-generation pipeline, we find the bound is never violated but badly loose: it over-predicts probe AUROC by 0.32–0.41 under exact-match/F1 labels (and 0.18–0.30 under an independent semantic judge, at re-selected layers).

The central question is how much of that gap is the estimator rather than the representation, and the answer is: most of it. Under a label-permutation null on the same activations – no class signal by construction – the estimator still returns a "certificate" of 0.87–0.94 for GPT-2 117M, GPT-2 Medium and Qwen2.5-0.5B, which against their same-layer probe AUROCs of 0.55–0.56 is 86–92% of the gap we report for those three models. Observed J exceeds its null at $p < 0.001$–0.045 uncorrected for four of six models, and is *not distinguishable from random labels at all* for the other two ($p = 0.195$, 0.380). Holding all 38 significance tests in the paper to one Holm-Bonferroni family leaves 3 surviving.

We also retract an invariance claim from an earlier version of this work: an apparent rotation-sensitivity of $\hat{J}$, which we had read as a basis-dependence of fixed-rank truncation, traces entirely to an unseeded randomized SVD and vanishes under a deterministic solver. We report the null control and the determinism check as the two diagnostics any Fisher-style detectability ceiling should carry – offered here because they overturned our own headline rather than confirmed it.

Keywords: Fisher separability, geometric certificates, hallucination detection, representation geometry, optimal transport, conformal prediction

1. Introduction

A *geometric certificate* asks whether a scalar computed from hidden-state statistics alone – no probe fitting, no optimisation – predicts the AUROC any linear detector could achieve at a given layer. Fisher separability J is the natural candidate: under Gaussian equal-covariance assumptions, the Bayes-optimal linear classifier achieves AUROC $= \Phi(\sqrt{J/2})$ (Proposition 1, Appendix D.1), the classical binormal ROC identity (Green and Swets, 1966; Bamber, 1975; Hanley and McNeil, 1982), restated for self-containedness and applied at scale to LLM hidden states rather than claimed as novel – a *diagnostic* use distinct from using J *operatively*, as a classifier itself (Ricco et al., 2026). As detailed below, the certificate is *training-free*, not *label-free* (J needs the same class-labelled centroids a probe would).

Central thesis. Detection performance is a property of representation geometry, not of the probe. This paper is a systematic, adversarially self-audited test of that thesis

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across six models, three labeling pipelines, and four certificate families (Fisher, Bures W_2, Sliced Wasserstein, MMD): every case where an initially promising pattern did not survive a matched-methodology or minority-class-size re-test is reported as such. The full ten-item contribution list and complete validation history are in Appendix A.

2. The Certificate and Its Central Result

Setup (Appendix C). Binary correct/hallucinated classification of LLM responses on HaluEval and TruthfulQA, via three methodologically distinct, non-interchangeable pipelines – A (forced-choice), B (real generation, ROUGE-L), C (real generation, SQuAD-style EM/F1) – that must not be pooled across models when comparing scale (a confound this paper's own earlier drafts made and then corrected). $J(L) = (\mu_c - \mu_h)^\top \Sigma_w^{-1} (\mu_c - \mu_h)$, estimated via PCA(100) + Ledoit–Wolf shrinkage (Appendix C.4).

Central result (Exp. 15, six models, Pipeline C, one matched methodology; full derivation history in Appendix D.2). Table 1 gives the result the paper's earlier, cross-pipeline validation lacked. Under one consistent methodology, the corrected identity $\Phi(\sqrt{J/2})$ over-predicts same-layer linear-probe AUROC by 0.32–0.41 at every one of six scales tested – an order of magnitude looser than the $\leq 2.7\%$ error the original, now-superseded validation reported by mixing pipelines across scale and measuring only models already at ≈0.99–1.00 AUROC, where a bound pinned near 1.0 trivially matches.

Table 1: Matched-methodology certificate validation (Exp. 15, Pipeline C; full table with Probe-AUROC and hallucination-rate columns is Table 6 in Appendix D.2).

Model	J	Bound	Same-Layer AUROC
GPT-2 (117M)	7.52	0.9737	0.564
GPT-2 Medium	3.97	0.9207	0.561
Qwen 2.5 0.5B	3.69	0.9129	0.551
Qwen 2.5 1.5B	2.25	0.8556	0.539
Qwen 2.5 3B	2.00	0.8412	0.504
Qwen 2.5 7B	1.63	0.8169	0.484

The same corrected experiment initially suggested a second finding – J decreasing perfectly monotonically with scale (Spearman $\rho = -1.0$) – interpretable as stronger models' errors being harder to separate geometrically. **A direct test, fixing every model's minority (correct) class at $n = 24$, makes this disappear** ($\rho = +0.54$, $p = 0.27$, sign reversed). We withdraw it: it was a small-sample estimation artifact of J's covariance estimator, not a capability-dependent geometric phenomenon. Hallucination rate's own decline with scale ($\rho = -0.943$, $p = 0.0048$) is a simple proportion, unaffected by this issue, and is the one scale-covariate finding from this experiment that survives. Matched-methodology best-layer probe AUROCs themselves are 0.53–0.65 – at or barely above chance – so the looseness above is measured against a task where linear probing is already close to uninformative.

3. Three Principled Limitations and Two Further Failure Modes

Full derivations, robustness checks, and every correction round are in Appendix H and F–G.

Label validity. ROUGE-L and GPT-4o labels disagree at Cohen's $\kappa = -0.010$ (Appendix C.2); a second, independent cross-family judge audit of this paper's own central Pipeline-C labels finds similarly weak agreement ($\kappa = 0.141$, Appendix H). Every AUROC in this paper conditions on a labeling criterion a semantic judge substantially disagrees with.

Task validity. A six-feature surface classifier with no hidden states achieves AUROC$= 0.9773 \pm 0.0060$ on Pipeline A, matching the 0.986–0.9998 range of this paper's own hidden-state probes there, since HaluEval's hallucinated answers run $\approx 5\times$ longer than correct ones (Appendix C.1). Every Pipeline-A number here risks reflecting answer-length separability, not hallucination detection.

Distribution shift. Split conformal guarantees collapse under OOD transfer to TruthfulQA for every model tested (AUROC $\to$ 0.485/0.384/0.556), though the near-equal violation factors are largely an artifact of near-degenerate OOD label rates (3–7 positives per 400), not scale-invariance. A genuinely non-degenerate test – Pipeline C on NQ-open – does *not* replicate the collapse (split-consistent bootstrap gap mean -0.0009, 95% CI $[-0.183, +0.179]$), read as "no catastrophic collapse" from a floor effect, not demonstrated robustness (Appendix G.2).

Non-Gaussian geometry. $J \neq W_2^2$ (mean relative error 1.71, Qwen; 0.77, GPT-2), but distribution-free certificates do not exploit this. An earlier version reported Bures W_2/SW_2 beating Fisher on Qwen and reversing on GPT-2; rerunning on matched Pipeline-C data across *three* architectures with bootstrap CIs **withdraws that** (Appendix F): on two of three, *no* certificate is distinguishable from zero, and only on Pythia do all four agree rather than compete. Peak depth fraction is independently non-universal across all six models (67–100%, Appendix E.2).

Causal read/write gap. A Fisher-LDA steering vector on GPT-2 117M shows no significant causal specificity against a random-direction control at any strength (McNemar $p = 1.000/0.453/1.000$); the null replicates at 14B and for a sparse SAE-feature intervention. Several configurations could not have reached significance regardless of outcome (Appendix K).

4. The Null Control: Most of the Reported Gap Is the Estimator

The bound error above is a difference between an estimate and a probe. It is only a statement about representation geometry if the estimate carries signal. We therefore ran the control this work previously lacked: a label-permutation null on the same cached activations, with the class labels randomly shuffled so that no class structure exists by construction, using a deterministic solver throughout.

Result 1: the null reproduces most of GPT-2 117M's PCA-rank curve. GPT-2 117M's observed bound error rises from 0.250 at k=25 to 0.434 at k=200; permuted labels alone yield 0.204 to 0.426, i.e. 82% (k=25) to 98% (k=200) of the observed value, falling short by 0.008–0.049 (Table 2). $\hat{J}_{\text{null}}$ roughly doubles as k doubles ($1.07 \to 2.28 \to 4.79 \to 10.79$), i.e. grows about linearly in the number of retained components, which is the behaviour plug-in Fisher/Mahalanobis statistics are known to exhibit at $n \ll d$ (Bickel and

Levina, 2004; Fan and Fan, 2008). At k=200 the null-implied bound error falls 0.008 short of the observed bound error.

Table 2: Label-permutation null, GPT-2 117M at its best-J layer (probe AUROC 0.564; 100 permutations per cell; deterministic solver). "Null bound error" is the bound implied by $\hat{J}_{\text{null}}$ minus the probe AUROC – the gap obtainable with *no class signal at all*. "Observed bound error" is GPT-2 117M's own bound from the unpermuted labels at the same layer, k and solver, minus the same probe AUROC (at k=100 this is 0.408; Table 6 reports 0.410 from the earlier, non-deterministic solver).

k	$\hat{J}_{\text{null}}$	Null bound	Null bound error	**Observed bound error**
25	1.07	0.768	0.204	0.250
50	2.28	0.857	0.293	0.342
100	4.79	0.939	0.375	0.408
200	10.79	0.990	0.426	0.434

Result 2: at k=100, most of the gap is null across models, and for one model all of it is. Comparing each model's observed $\hat{J}$ against its own permutation null at the paper's own k=100 (200 permutations each): GPT-2 117M p=0.020, GPT-2 Medium p<0.001, Qwen 0.5B p=0.045, Qwen 3B p=0.025 – and for **Qwen 1.5B (p=0.380) and Qwen 7B (p=0.195) the certificate is not distinguishable from random labels at all**. Expressed as bound error, for GPT-2 117M, GPT-2 Medium and Qwen 0.5B the null accounts for 86–92% of what we report. The residual above-null signal is the only part attributable to the representations; it is marginal at best for four of six models and absent for two, and none of these uncorrected p-values survives the paper's Holm-Bonferroni family.

What we conclude. We do not withdraw the observation that the certificate is loose – it is, and it is never violated. We withdraw the implicit reading that the *size* of that looseness, or its dependence on k, is informative about hallucination geometry. Both are substantially properties of a plug-in estimator applied with n=500 in d up to 3584 with a minority class of 24–103. A detectability ceiling reported without a permutation null is, at these sample sizes, not yet evidence about the representation.

5. Conclusion

What survives this audit is narrower than what it set out to test. The Fisher bound is never violated across 91+ layer–model pairs, but a permutation null shows that for GPT-2 117M, GPT-2 Medium and Qwen2.5-0.5B 86–92% of its apparent looseness is reproduced with random labels, so the headline gap is largely an estimator-bias measurement at $n \ll d$ rather than a statement about representation geometry. An invariance claim from an earlier version is retracted outright (§D.3): it was an artifact of an unseeded solver. What we can still offer is methodological – a null control and a determinism check that any Fisher-style detectability ceiling should carry, and which we did not carry until they overturned our own result.

Guide to the Appendix

The four-page main text is a summary; it makes 29 cross-references into the sections below, which carry the evidence for every number it states. Reviewers are not expected to read all of it. The load-bearing sections are:

- **§D.2** – the central result (Table 6), the judge-relabelled re-test, the third-architecture extension, and the PCA-rank and probe-hyperparameter sensitivity analyses. Start here.
- **§D.3** – the basis-invariance analysis that explains *why* the PCA-rank sensitivity exists, with the nine-model spectral-gap test.
- **§F** – the optimal-transport comparison, including the withdrawal of a previously headline claim.
- **§H** – limitations, and the multiple-comparison correction across all 38 tests in the paper.

The remaining sections are supporting material: background and setup (§A–§C), the derivation (§D.1), scale behaviour (§E.2), conformal coverage (§G), the causal-intervention protocol (§K), and a correction history documenting every claim this paper revised or withdrew during its own review rounds. Code and result files are in the supplementary material.

Appendix A. Extended Introduction, Full Contribution List, and Terminology Notes

Fluent text and factual correctness are distinct properties of language model output. The hallucination phenomenon exploits this mismatch. Intercepting hallucinated responses before they reach a user is a prerequisite for safe deployment. Yet prevailing detection methods depend on labelled training data, a supervised probe, or repeated sampling (Manakul et al., 2023; Chen et al., 2024).

A qualitatively different approach is a *geometry-based certificate*: a scalar computable from hidden-state statistics alone. It predicts, before any probe is *fit* – no classifier training, no iterative optimisation – whether a linear detector can achieve a target AUROC. Such a certificate would recast hallucination detection as a representation-level property, assessable without training a classifier. **A note on terminology (added this revision, matching the paper's title):** we retain *certificate* throughout as the established name for this quantity. The paper's own findings (§D.2, §G.2) show it functions as a *loose ceiling*, not a tight guarantee. It over-predicts achievable AUROC by 0.32–0.41 at the best-matched layer, and the probes it bounds are themselves near-chance (0.53–0.65 best-layer AUROC) under the paper's most careful labeling pipeline. Read *certificate* below as shorthand for this bound – not as a claim that it is tight, causal, or safe to deploy as an operational gate.

Correction, added post-review (second round): this is *training-free*, not *label-free*. An earlier draft described the certificate as usable "without ground-truth labels." This contradicts its own definition below: J requires the class centroids μ_c, μ_h, computed from the same labelled correct/hallucinated examples a probe would use. The certificate

saves the cost of fitting and cross-validating a classifier. It does not remove the labelling requirement a probe already has. Any claim in this paper of deployment without labels refers only to skipping the classifier-fitting step, not to skipping labelled data entirely.

Fisher separability J is the natural candidate. Under Gaussian equal-covariance assumptions, the maximum AUROC achievable by any linear classifier on two class distributions equals $\Phi(\sqrt{J/2})$, where Φ denotes the standard normal CDF. J depends only on the first two moments of the hidden-state distribution at each layer. Beyond the class centroids and pooled covariance it is computed from, it requires no *further* labelled data or optimisation to produce a bound – unlike a probe, which additionally needs labels for cross-validated model selection.

Certificate vs. classifier. This framing is categorically distinct from using Fisher geometry as a decision boundary. Concurrent work (Ricco et al., 2026) applies Fisher separability as a classifier, to detect hallucinations in small-model embedding space, achieving F1 $>$ 90%. The contribution here is orthogonal. Rather than committing to a specific Fisher-based detector, we ask what AUROC ceiling the geometry imposes on *any* linear classifier at a given layer – before a single training example is used. A certificate is *diagnostic*: it characterises whether a representation supports detection. A classifier is *operative*: it exploits that support. Table 3 clarifies what this certificate provides and what it does not.

Table 3: Scope of the geometric certificate developed in this paper.

Property	**Provided**	**Not provided**
Upper-bound linear probe AUROC (valid direction, every case)	✓	
Predict linear probe AUROC *tightly* under one matched, real-generation pipeline (error 0.32–0.41, Exp. 15)		×
Predict linear probe AUROC tightly in saturated/near-degenerate regimes only ($\leq$ 2.7% error, Exp. 01/06/13 – not representative, see §D.2)	✓	
Bound shallow MLP probe empirically (0 violations, 91 pairs, saturated regime only; not retested under matched methodology)	✓	
Scale to 7.6B parameters, matched methodology (Qwen 2.5 7B, Exp. 13, 15)	✓	
Scale to 14.8B parameters, Pipeline A only, not matched (Qwen 2.5 14B, Exp. 20)	✓	
Select the optimal probing layer		×
Bound OOD (out-of-distribution) performance		×
Formal in-distribution coverage ($\alpha^* = 0.04$–0.07)	✓	
Factual hallucination detection		×
Pipeline A (≈saturated, forced-choice) numbers as evidence *about* hallucination geometry, independent of answer-length		×

Added, final-audit pass: read every Pipeline A number in this paper as certificate/probe math, not as hallucination-detection evidence. Exp. 18 (§D.2) shows Pipeline A's near-saturated AUROCs are statistically indistinguishable from a 6-feature answer-length classifier that never touches a hidden state (AUROC 0.9773 ± 0.0060, permutation $p = 0.0020$, matching the 0.9859–0.9998 range of the actual hidden-state probes on that same pipeline). Every table in this paper that includes a Pipeline A row (Tables 5, 7, 9) is flagged inline at first use, but we state it once, prominently, here as well: Pipeline A rows validate the certificate's *mathematical* bound (does the AUROC ceiling ever get violated) under a task any answer-length classifier already solves near-perfectly, not the certificate's *relevance* to detecting hallucination content. Table 6 (Pipeline C, matched real-generation methodology, identical across all six models) is this paper's primary evidence on that latter question and should be read as such. **Elite-review follow-up, disclosed as an open gap rather than silently left: this same surface-only confound check has never been run on Pipeline C.** Pipeline C scores real, free-form generations by SQuAD-style exact-match/F1 against a reference answer – a metric with its own length- and verbosity-sensitivity – and its best-layer probe AUROCs are themselves only 0.53–0.65, an order of magnitude closer to chance than Pipeline A's near-saturation, so a surface-length or word-overlap classifier could plausibly explain a substantial share of that residual signal. We do not have this control and do not claim Table 6 is free of it; a length/verbosity-only baseline on Pipeline C's own generations, exactly analogous to Exp. 18, is the concrete next step this gap motivates.

This paper makes the following contributions:

1. We apply the classical Gaussian binormal certificate $\Phi(\sqrt{J/2})$ (Green and Swets, 1966; Bamber, 1975; Hanley and McNeil, 1982) to LLM hidden-state hallucination detection. We validate it empirically at scale, across six models, under one consistent real-generation methodology (Exp. 15). Bound error is 0.32–0.41 at every scale tested – an order of magnitude looser than the $\leq 2.7\%$ maximum / 1.2–1.4% mean error our original, cross-pipeline-confounded validation reported for Qwen 2.5 3B and GPT-2 Medium (Exp. 01, 06). **The certificate is not empirically tight** once tested outside saturated or near-degenerate label regimes – a correction to this paper's original headline claim.

2. Under the matched methodology, Fisher separability J initially appeared to decrease perfectly monotonically with model scale, across all six models tested (Spearman $\rho = -1.0$). **A direct subsampling test – fixing every model's minority-class size at $n = 24$ – shows this was a small-sample estimation artifact, not a genuine capability-dependent geometric phenomenon** (matched-n Spearman $\rho = +0.54$, $p = 0.27$, sign reversed). We withdraw this claim. Hallucination rate's own significant decline with scale ($\rho = -0.943$, $p = 0.0048$) is a simple proportion, unaffected by the covariance-estimation issue. It is the one scale-covariate finding from Exp. 15 that survives.

3. In the original (now-superseded) saturated-regime validation, we show via 5-fold cross-validation that the certificate predicts unseen-fold AUROC within 0.93% mean error on Qwen 2.5 3B, before any probe is trained (Exp. 03). This is subject to the same saturation caveat as contribution 1.

4. We identify a fundamental limitation: Fisher J cannot select the optimal probing layer (0% argmax match rate). The certificate bounds achievable performance but not where to find it (Exp. 03).
5. We characterize the non-Gaussian geometry: $J \approx W_2^2$ only approximately (mean relative error 1.71 for Qwen). Bures W_2 ($\rho = 0.828$) and SW_2 ($\rho = 0.821$) both substantially outperform Fisher ($\rho = 0.458$) on Qwen, but reverse entirely on GPT-2. This establishes architecture-dependent certificate selection (Exp. 08).
6. We derive and validate a split conformal guarantee ($\alpha^* = 0.07$, Qwen; $\alpha^* = 0.06$, GPT-2). We show it collapses under out-of-distribution shift (Exp. 10).
7. We document that the task itself is ambiguous: ROUGE-L and GPT-4o labels disagree at $\kappa = -0.010$ (Exp. 07). This scopes all results to surface-form divergence.
8. We show that the bound holds empirically for shallow MLP probes as well as linear ones. Zero violations occur across all 91 layer–model pairs; MLP gain is $\leq 0.09\%$ over linear (Exp. 12).
9. We extend the full certificate validation to Qwen 2.5 7B (7.6B parameters, $d = 3584$, 29 layers). Zero violations occur across all 29 layer-model pairs; MLP gain is < 0.0001; mean bound error is 0.05%; and the conformal guarantee is $\alpha^* = 0.04$ (52.8% acceptance rate), on the same balanced HaluEval setup (Exp. 13).
10. **Added post-review (round 6): a further scale extension to Qwen 2.5 14B (14.8B parameters, $d = 5120$, 49 layers, official pre-quantized 4-bit AWQ checkpoint) on the identical Pipeline-A protocol (Exp. 20).** Zero violations occur across all 49 layer-model pairs; the certificate holds for both linear and MLP probes. Best-layer linear AUROC is 0.9909 (L29); MLP AUROC is 0.9897 (L25, a -0.12pp gain – the MLP does not beat the linear probe at this scale). Mean/max bound error is 0.0145/0.0318 at the best-J layer (L30, same-layer comparison). The conformal guarantee is $\alpha^* = 0.01$ (53% acceptance rate, empirical hallucination rate 0.94% among accepted queries). This result required four earlier attempts at a larger, 32B extension that failed on Kaggle GPU-memory-placement bugs specific to on-the-fly bitsandbytes 4-bit quantization (documented in the experiment script). Switching to an officially pre-quantized AWQ checkpoint resolved it. The qualitative picture from Qwen 7B is unchanged at 14B: the certificate holds, remains loose by a similar margin, and the MLP-over-linear gain remains negligible-to-negative.
11. We confirm, *qualitatively*, that OOD guarantee collapse is not resolved by scale within one model family. Both Qwen 2.5 3B and Qwen 2.5 7B (Exp. 10, 14) produce conformal guarantees that provide no usable coverage on TruthfulQA. **Correction (round 2):** the originally-reported near-equal violation factors (14.0× vs. 14.2×) are largely an arithmetic artifact. Both are computed by dividing two similarly near-degenerate TruthfulQA hallucination rates (on just 3–7 positive examples out of 400) by two similar in-distribution α^* values. This is compounded by a task confound between the in-distribution (forced-choice) and OOD (real-generation) pipelines (§G.2). We withdraw the quantitative scale-invariance reading and retain only the qualitative one. Whether this extends across architectures remains untested: no genuine cross-architecture real-OOD experiment existed in this study before this revision. **Closed, post-review**

(fourth round): we ran the previously-missing GPT-2 Medium data point, using hidden-state tensors already cached from earlier extractions. It reproduces the same near-degenerate OOD label problem (4/400 positive) as both Qwen models, at a comparable violation factor (14.1×) – evidence that the artifact, not the collapse magnitude, generalises across architectures (§G.2). **Disambiguation (round-5 review):** "matched" here means matched to the same in-dist(Pipeline A)/OOD(Pipeline B) protocol used for Qwen, not matched to Exp. 15's Pipeline C. An earlier draft's "methodology-matched" phrasing conflated the two senses. Whether the collapse *magnitude* itself is architecture-invariant remains untested, since no OOD labeling pipeline in this study yields a non-degenerate positive class on TruthfulQA.

Central thesis. Detection performance is a property of representation geometry, not of the probe. Fisher separability is a classical discriminant criterion. What has not been systematically examined is whether it constitutes a *predictive, training-free certificate* for linear probe performance, specifically in large language models. This paper provides that examination: quantitative validation across architectures, failure-mode characterisation, and a formal coverage guarantee. The classifier is, in this framing, merely a *measurement instrument* reading a signal already encoded in the geometry.

Appendix B. Background

B.1. Fisher Discriminant and Linear Separability

Let $\mathcal{H}_c, \mathcal{H}_h \subset \mathbb{R}^d$ denote the sets of hidden states for correct and hallucinated responses at a given layer L. Define class centroids $\mu_c, \mu_h \in \mathbb{R}^d$ and the pooled within-class covariance $\Sigma_w = \frac{n_c\Sigma_c + n_h\Sigma_h}{n_c+n_h}$. We assume a shared pooled covariance Σ_w across classes, consistent with classical Fisher linear discriminant analysis (Fisher, 1936). Class-specific covariances Σ_c, Σ_h enter only through this pool. The Fisher ratio measures class separation in this covariance-weighted metric:

$$J(L) = (\mu_c - \mu_h)^\top \Sigma_w^{-1} (\mu_c - \mu_h). \tag{1}$$

Under the Gaussian equal-covariance model $P(h \mid y) \sim \mathcal{N}(\mu_y, \Sigma_w)$, the Bayes-optimal linear classifier on the Mahalanobis-projected line achieves

$$\text{AUROC} = \Phi\left(\sqrt{\frac{J}{2}}\right), \tag{2}$$

where Φ is the standard normal CDF. This is the classical binormal ROC identity from signal detection theory (Green and Swets, 1966; Bamber, 1975; Hanley and McNeil, 1982). It relates the area under an ROC curve to the standardised separation between two equal-variance Gaussians. We make no claim to originality for the identity itself. Our contribution is its application, at scale, to LLM hidden-state hallucination detection. We use it as a training-free characterisation of the detection ceiling a representation permits, rather than as a classifier. We then measure empirically where, and how tightly, it holds across architectures, layers, and model scale. The formula arises from projecting onto the Fisher direction. The resulting score is Gaussian under each class; the standardised mean gap equals $\sqrt{J}$; and AUROC reduces to the area under the resulting normal curves. We make this derivation explicit in Proposition 1 for completeness, following the textbook argument.

We distinguish two senses throughout. *Theoretically*, (2) is an *approximation*. The Gaussian equal-covariance assumption is violated in practice, so the formula may under- or over-predict AUROC. *Empirically*, we observe that $\Phi(\sqrt{J/2})$ consistently *over-predicts* AUROC across all tested layers and models. It functions as an empirical upper bound, even though no theoretical guarantee of this direction exists. We use "bound" to refer to this empirical behaviour, and "approximation" when speaking of the Gaussian assumption itself.

B.2. Optimal Transport and the Gaussian Limit

The 2-Wasserstein distance between distributions P, Q is $W_2(P,Q) = \inf_{\gamma \in \Pi(P,Q)} \left(\mathbb{E}_{(x,y)\sim\gamma} \|x - y\|^2\right)^{1/2}$. When both distributions are Gaussian with equal covariance Σ_w, the whitened distance satisfies $W_2^2 = J$, making Fisher ratio the *Gaussian special case* of optimal transport (Sugiyama, 2007). Sliced Wasserstein distance (SW_2) approximates W_2 via random one-dimensional projections. It makes no Gaussianity assumption, providing a distribution-free alternative certificate (Kolouri et al., 2019).

B.3. Conformal Prediction

Split conformal prediction (Angelopoulos and Bates, 2022) offers a distribution-free path from a trained scorer to a formal coverage guarantee, without distributional assumptions. The mechanism is threshold calibration. Holding out a calibration set, we identify the smallest rejection threshold τ such that the fraction of calibration points with hallucination score above τ is at most α. By exchangeability, this threshold then satisfies $P(\text{hallucinated} \mid \text{accepted}) \leq \alpha$ on fresh test data, where ACCEPT is defined as score $\leq \tau$.

Appendix C. Experimental Setup

C.1. Task and Datasets

We study binary classification of LLM responses as *correct* or *hallucinated*. All experiments use two datasets, extracted via *three methodologically distinct pipelines*. This paper's first two review rounds required us to disentangle these explicitly (see the correction below): they are not interchangeable, and the label "ROUGE-L" does not describe all of them.

HaluEval (Li et al., 2023): 2,000 question–answer pairs (in-distribution). **Pipeline A (forced-choice, no generation, no ROUGE-L)** covers Exps. 00, 01, 03, 08, 09, 10, 13 – Qwen 2.5 3B, GPT-2 Medium, and Qwen 2.5 7B. Here the model is forward-passed directly on the pre-written reference text (`"Q: {question}\nA: {right_answer}"` → label 1, or the pre-written `hallucinated_answer` → label 0). No text is generated and no ROUGE-L threshold is applied. Labels are balanced by construction (1000/1000 pairs, empirically 497/497 confident splits in the conformal analysis of §G.2). This is an easier task than detecting a model's own hallucinated generations: the model only represents text it did not produce, some of which is already labelled correct or incorrect by dataset construction.

Item-difficulty / virtual-circuit caveat, added post-review (fourth round), then confirmed quantitatively (round-5 review): Pipeline A's near-saturated AUROCs (≈ 0.99) may in part reflect a *lexical or template-identity* distinction – which of two pre-written strings was fed in – rather than hallucination detection in any meaningful sense. **We ran the control that dissociates the two, and the result is not reassuring.** A trivial surface-feature classifier uses six hand-built features of the answer string alone: character length, word count, Jaccard word-overlap with the question, comma count, initial-capitalization, question length. *It has no hidden states and makes no model forward pass at all* (`experiments/18_pipeline_a_surface_confound_check.py`, run on 1000 HaluEval QA pairs re-loaded directly from `pminervini/HaluEval`). It achieves 5-fold CV AUROC $= 0.9773 \pm 0.0060$ (permutation test, 500 shuffles, $p = 0.0020$), essentially matching the 0.9859–0.9998 range of the actual hidden-state probes on this same pipeline. The confound is not subtle. Mean answer length is 13.6 characters / 2.2 words for `right_answer`, versus 64.5 characters / 10.7 words for `hallucinated_answer`. HaluEval's hallucinated answers are, on average, nearly 5× longer than the correct ones – a systematic dataset-construction artifact, not a property of truthfulness. **This means every Pipeline-A number in this paper (Exps. 00, 01, 03, 08, 09, 10, 13; Tables 9, 7, 5, and the OOD in-distribution arm of §G.2) is at serious, now-quantified risk of reflecting answer-length/verbosity separability rather than hallucination detection.** We can no longer treat "detecting which string was shown" and "detecting whether the content is true" as merely undissociated: the surface-only control shows they are close to *indistinguishable in this dataset at the AUROC level*. We do not retract the Pipeline-A numbers – the hidden-state probes still do something, and their small residual advantage over the pure surface classifier, where present, is worth noting layer-by-layer in future work. But any claim resting primarily on Pipeline-A separability should be read as a claim about a confounded task, not a clean measurement of hallucination detectability.

TruthfulQA (Lin et al., 2022): 400 questions. **Pipeline B (real generation, ROUGE-L ≥ 0.4)** covers the Qwen 2.5 0.5B/1.5B rows of the scale curve (Exp. 11) and the §G.2 OOD-collapse experiment (Exp. 10, real `real_ood_truthfulqa` block; Exp. 14). Here the

model actually generates a response to the bare question, and the response is scored against the reference answer via ROUGE-L. This is the pipeline the $\kappa = -0.010$ audit below applies to.

Pipeline C (real generation, SQuAD-style EM/F1) was added in this revision (Exp. 15). It gives the six models compared in Table 5 and the scale curve a single, matched methodology: real generation given only the question, scored via normalized exact-match/token-F1 against the HaluEval reference answer, identical to the labeling fix independently validated in a companion paper's agentic-failure taxonomy. See the correction in §D.2 and §E.1 for how this changes the in-distribution comparison.

C.2. A Note on Label Validity

For Pipeline B (real generation, ROUGE-L threshold), we audited the labeling choice. We re-labelled 700 HaluEval responses using GPT-4o as a judge (Exp. 07). The agreement between ROUGE-L and GPT-4o labels is Cohen's $\kappa = -0.010$ (ROUGE-L hallucination rate = 0.993; GPT-4o rate = 0.981). Near-zero κ indicates the two labelling strategies identify *different* samples as hallucinated, despite similar rates. **Caveat, added post-review (fourth round):** at ≈99% prevalence, Cohen's κ is dominated by a small minority of examples. It is a statistically unstable measure of agreement – a handful of label flips in either direction can move κ substantially. We report $\kappa = -0.010$ as evidence of qualitative disagreement between the two labeling strategies (which samples each flags), not as a precise, low-variance estimate of agreement strength. A raw percent-agreement figure, or a bootstrapped CI on κ, would be needed before this number should be read as more than a directional signal. This audit does *not* extend to Pipeline A, whose labels are definitional – correct/incorrect by dataset construction, not derived from any text-similarity metric. Nor was it re-run for Pipeline C's EM/F1 scoring; both remain unaudited against an LLM-judge.

> **Scope condition (applies throughout, corrected in this revision):** *Results in this paper condition on* one of three distinct labeling pipelines*: A (forced-choice/definitional), B (real generation scored by ROUGE-L), or C (real generation scored by SQuAD EM/F1). They do not condition uniformly on ROUGE-L labels, as an earlier draft incorrectly stated. Every table and figure in this paper is now annotated with which pipeline produced it. None of the three establishes factual correctness in the strict sense. Pipeline B's $\kappa = -0.010$ gap to an LLM judge is the only one of the three with a direct audit against a stronger label source. Any claim about "hallucination detection" using Pipeline A or B data is shorthand for "surface-form or template divergence detection," not verified factual correctness.*

C.3. Hidden-State Extraction

For each input, we extract the last-token hidden state $h_L \in \mathbb{R}^d$ at every layer $L = 0, \ldots, L_{\max}$. Last-token position is used throughout, consistent with prior work (Azaria and Mitchell, 2023; Du et al., 2024). Table 4 lists the models used.

Table 4: Models and extraction parameters.

Model	**Params**	d	L	n
GPT-2 (117M)	117M	768	13	2000
GPT-2 Medium	345M	1024	25	2000
Qwen 2.5 0.5B	494M	896	25	400
Qwen 2.5 1.5B	1.54B	1536	29	400
Qwen 2.5 3B	3.09B	2048	37	2000
Qwen 2.5 7B	7.6B	3584	29	2000

C.4. Fisher Separability Computation

Direct inversion of $\Sigma_w \in \mathbb{R}^{d\times d}$ is numerically unstable at $d = 2048$. We apply two complementary approximations: (i) **PCA projection** to $k = 100$ principal components (capturing $> 95\%$ of variance in practice); and (ii) **Ledoit–Wolf shrinkage** (Ledoit and Wolf, 2004) for covariance regularization. Both methods produce consistent J estimates; PCA projection is used throughout. For the Qwen 0.5B and 1.5B models at $n = 400$, the aspect ratio $\gamma = d/n > 1$ (values 2.24 and 3.84 respectively), making full-rank covariance estimation unreliable. These results should be interpreted accordingly.

Appendix D. The Geometric Certificate

D.1. Derivation

Proposition 1 (Gaussian AUROC Certificate; classical result, restated for our setting) *Let $P_c = \mathcal{N}(\mu_c, \Sigma_w)$ and $P_h = \mathcal{N}(\mu_h, \Sigma_w)$ with equal covariance Σ_w. The AUROC of the Bayes-optimal linear classifier equals $\Phi\left(\sqrt{\frac{J}{2}}\right)$ where $J = (\mu_c - \mu_h)^\top \Sigma_w^{-1}(\mu_c - \mu_h)$. This is the binormal-model ROC identity of Green and Swets (1966); Bamber (1975); Hanley and McNeil (1982); we restate and prove it here only for self-containedness, not as a novel result.*

Proof [Derivation under Gaussian assumptions] Project onto the Fisher direction $w = \Sigma_w^{-1}(\mu_c - \mu_h)$. The resulting scalar score $s(h) = w^\top h$ is Gaussian under each class: $s \mid P_c \sim \mathcal{N}(\mu_c^\top w,\, w^\top \Sigma_w w)$ and $s \mid P_h \sim \mathcal{N}(\mu_h^\top w,\, w^\top \Sigma_w w)$. Substituting w, the common variance is $\sigma^2 := w^\top \Sigma_w w = J$ and the mean difference is $\mu_c^\top w - \mu_h^\top w = J$ as well, giving the standard signal-detection-theory effect size $d' := (\mu_c^\top w - \mu_h^\top w)/\sigma = J/\sqrt{J} = \sqrt{J}$. AUROC equals $P(s_c > s_h)$ for independent draws $s_c \sim \mathcal{N}(\mu_c^\top w, \sigma^2)$, $s_h \sim \mathcal{N}(\mu_h^\top w, \sigma^2)$; the difference $D = s_c - s_h$ is Gaussian with mean J and variance $2\sigma^2 = 2J$ (variances add for independent variables), so $\text{AUROC} = P(D > 0) = \Phi\left(\frac{J}{\sqrt{2J}}\right) = \Phi\left(\frac{\sqrt{J}}{\sqrt{2}}\right) = \Phi\left(\sqrt{\frac{J}{2}}\right) = \Phi(d'/\sqrt{2})$, the standard binormal-model identity. (An earlier draft skipped the $D = s_c - s_h$ step and conflated $d'/2$ – the argument of the equal-prior *Bayes accuracy* $1 - \Phi(-d'/2)$ – with the AUROC argument $d'/\sqrt{2}$; the two coincide only if $2 = \sqrt{2}$, which they do not.) ■

Remark 2 *Proposition 1 assumes Gaussian equal-covariance distributions. Real hidden states violate both assumptions. Eq. (2) therefore serves as an approximation, whose tightness we measure empirically.*

D.2. Empirical Validation

We compute $J(L)$ at every layer and compare $\Phi(\sqrt{J/2})$ to the 5-fold cross-validated linear probe AUROC. Table 5 reports key results.

Correction, added post-review (second round): the rows above are drawn from three different labeling pipelines (§C.1). They were previously presented as if directly comparable. GPT-2 (117M) and Qwen 0.5B use real generation scored against reference text (Pipeline B). GPT-2 Medium, Qwen 3B, and Qwen 7B use forced-choice forward passes on pre-written reference/hallucinated text with no generation at all (Pipeline A) – an easier task by construction, since the model never has to produce its own (possibly wrong) answer. This confound is corrected by Exp. 15 (Table 6), which reruns all six models under one real-generation, SQuAD-EM/F1 pipeline.

§**Correction, added post-review (second round):** the GPT-2 Medium row previously reported AUROC = 0.9887 at L19. This is actually the probe AUROC at **L8** (the probe-best layer, `01_fisher_analysis.json`: `best_probe_layer=8`, `best_probe_auroc=0.988679`) – a transcription error mixing two different layers' numbers into one row. L19 is the Fisher-J-best layer this table is supposed to report (`best_J_layer=19`); its actual probe AUROC

Table 5: Certificate validation at best-J layer (Exp. 01, 06, 13, 20). †GPT-2 117M: AUROC from prior MECH-INT extraction, own generation-based pipeline; J not recomputed here. **Final-audit note:** this specific value (0.604, logged as "MECH-INT Step 4C" in `results/logs/01_fisher_analysis.json`) could not be independently re-verified against any of the twelve saved MECH-INT result artifacts checked during this audit. The closest match is the dense-probe layer-9 AUROC (0.5827), which is close but not equal. We flag this as an unverified transcribed number, pending a fresh extraction pass, rather than silently treat it as confirmed. **Pipeline column added post-review**: rows are *not* methodologically comparable to each other. Pipeline A (forced-choice, no generation) vs. Pipeline B (real generation, ROUGE-L) are different tasks (see §C.1). ‡Qwen 7B: Exp. 13, Pipeline A. ¶Qwen 14B (Exp. 20, added post-review, round 6): official pre-quantized 4-bit AWQ checkpoint (`Qwen2.5-14B-Instruct-AWQ`), same Pipeline-A protocol, $n = 1000$ balanced vectors. This is a genuine scale extension (7.6B → 14.8B), obtained after four earlier attempts at a 32B extension failed on repeated Kaggle GPU-memory-placement bugs with on-the-fly bitsandbytes quantization, documented in `kaggle/exp20_notebook/exp20_qwen32b_scale_extension.py`. Mean/max errors are across all layers, same pipeline as that row. A matched-methodology replacement (Pipeline C, real generation + SQuAD EM/F1, identical across all six models) is reported in Table 6 where available.

Model	Pipeline	Layer	Depth	J	Bound	AUROC	Error
GPT-2 (117M)†	B (MECH-INT)	L9	69%	—	—	0.604	—
Qwen 2.5 0.5B	**B (Exp. 11)**	**L7**	**29%**	**24.5**	**0.9998**	**0.8135**	**0.186**
GPT-2 Medium	A (Exp. 01)	L19	79%	49.3	1.0000	0.9855§	0.014
Qwen 2.5 3B	A (Exp. 01)	L25	69%	51.9	1.0000	0.9917	0.008
Qwen 2.5 7B‡	A (Exp. 13)	L15	54%	184.7	≈1.000	0.9998	<0.001
Qwen 2.5 14B¶	A (Exp. 20)	L30	62%	57.8	≈1.000	0.9894	0.011
All layers, Qwen 3B + GPT-2 Med (Pipeline A):							
Qwen 3B — max / mean error							0.027 / 0.012
GPT-2 Med — max / mean error							0.021 / 0.014
Qwen 7B — max / mean error							0.004 / 0.0005
Qwen 14B — max / mean error							0.032 / 0.014

is 0.9855 (`probe_auroc_per_layer[19]`), giving a corrected error of 0.014, not 0.011. This is exactly the layer-selection failure documented in §D.5: Fisher J and probe AUROC peak at different layers, L19 vs. L8. The corrected number now correctly *illustrates* that failure, instead of accidentally hiding it behind the wrong layer's AUROC.

Correction, added post-review (second round): an earlier revision of this paper computed the binormal AUROC identity as $\Phi(\sqrt{J}/2)$. The correct identity is $\Phi(\sqrt{J/2})$, following the standard signal-detection-theory derivation: projecting onto the Fisher direction gives a standardised separation $d' = \sqrt{J}$, and AUROC $= \Phi(d'/\sqrt{2})$ for two equal-variance Gaussians. The two differ because $1/2 \neq 1/\sqrt{2}$; the wrong formula silently computed the equal-prior *Bayes accuracy*, not the AUROC. All bound values, table entries, and figures in this paper now use the corrected formula (Proposition 1, §D.1); `src/fisher.py:auroc_bound` carries a permanent note on the fix.

Correcting the formula exposes a more serious problem than the one flagged in the first review round. The "$\leq$ 2.7% maximum error" headline figure quoted in the abstract is computed only over Qwen 3B, GPT-2 Medium, and Qwen 7B – three models whose best-layer AUROC is already ≈0.99–1.00. A bound pinned near 1.0 trivially matches an AUROC pinned near 1.0. The one model in this study with a genuinely non-saturated measured AUROC is **Qwen 2.5 0.5B** (best AUROC 0.8135). Under the corrected formula its predicted bound is **0.9998** – *indistinguishable, to four decimal places, from every other model's bound in this table* – despite Qwen 0.5B's true AUROC sitting 18.6 points lower than the others. This is not a coincidence of one non-saturated data point being slightly looser than the rest. Every single J value measured anywhere in this study ($J \geq 24.5$) is already large enough that $\Phi(\sqrt{J/2})$ saturates to ≈0.9998–1.000, regardless of the model's true achievable AUROC. The certificate, as validated here, predicts "≈1.0" unconditionally. It has never been tested in the only regime where a bound would be informative: a layer/model whose true J is small enough to keep $\Phi(\sqrt{J/2})$ meaningfully below 1. The bound error of **0.186** for Qwen 0.5B – roughly 7× *larger* than the worst per-layer error reported anywhere else in this table – should be read as the honest headline result of Table 5. It is not evidence the certificate is occasionally loose. It is evidence the certificate has not yet been exercised outside the saturating regime.

Across the four saturated layer–model combinations counted separately from Qwen 0.5B (91 layer–model pairs total: 37 + 25 + 29 across GPT-2 Medium/Qwen 3B/Qwen 7B), the bound error never exceeds 0.027. The bound is a valid upper approximation throughout: $\Phi(\sqrt{J/2}) \geq \text{AUROC}_{\text{actual}}$ holds in every case, consistent with the Gaussian assumption overestimating separation. At 7B scale under this pipeline ($J \approx 185$, the Pipeline-A forced-choice regime), the Fisher bound saturates to ≈ 1.000 and mean error drops to 0.05%. **Correction, added post-review (third round): this sentence must not be read as "the certificate becomes exact at scale" in general.** It describes only this specific saturated pipeline. The same Qwen 2.5 7B model, under the matched Exp. 15 methodology (Table 6), gives $J \approx 1.63$ – two orders of magnitude smaller – and a bound error of 0.332: the certificate is loose, not exact, for this model once the task is not artificially easy. The $J \approx 185$ value is a property of the forced-choice task, not of the model's representations becoming more separable in any general sense.

Closed, elite-review follow-up: an adversarial reviewer flagged that the Error column could not be reconstructed from this table's own adjacent visible

Table 6: **Exp. 15 – Pipeline C, matched methodology (this paper's primary certificate-validation table).** Real generation given only the question (500 HaluEval questions per model, no reference text shown to the model), scored via SQuAD-style EM/F1 against the reference answer. The pipeline and labeling scheme are identical across all six models. Layer/depth and J/Bound are at the best-J layer; Probe AUROC is the best value over all layers (not necessarily the same layer as J); Same-Layer AUROC is the linear-probe AUROC at that same best-J layer, added here (elite-review follow-up) because it is the number **Error** is actually computed from – $|\Phi(\sqrt{J/2}) - \text{Same-Layer AUROC}|$ – not the adjacent Probe AUROC column, and it does not appear anywhere else in this table. **Same-Layer AUROC is below the Probe AUROC column at every one of the six layers, and below chance for Qwen 2.5 3B and 7B (0.504 and 0.484).** This is corrected, post-third-round-review, from an earlier version of this table, which subtracted the best-*probe*-layer AUROC from the best-J-layer bound – two different layers – understating the looseness; the same-layer figures used here are larger, and now shown directly rather than left reconstructable only from the raw logs. Unlike Table 5, every row here comes from the same task. The pattern across rows is therefore a genuine scale comparison, not an artifact of mixing forced-choice and generation pipelines.

Model	Depth	J	Bound	Probe AUROC	Same-Layer AUROC	Error	Hall. rate
GPT-2 (117M)	67%	7.52	0.9737	0.6548	0.564	0.410	0.952
GPT-2 Medium	83%	3.97	0.9207	0.6081	0.561	0.360	0.900
Qwen 2.5 0.5B	100%	3.69	0.9129	0.5840	0.551	0.362	0.906
Qwen 2.5 1.5B	79%	2.25	0.8556	0.6022	0.539	0.316	0.880
Qwen 2.5 3B	100%	2.00	0.8412	0.5719	0.504	0.337	0.810
Qwen 2.5 7B	86%	1.63	0.8169	0.5254	0.484	0.332	0.794

columns, and that the underlying same-layer AUROC values – as low as 0.484, below chance, for Qwen 2.5 7B – were never shown anywhere. The caption already stated Error uses a same-layer AUROC rather than the Probe AUROC column, but that same-layer number itself was absent from the table, making the claim unverifiable by a reader and hiding the fact that the certificate's own diagnostic layer is, for the two largest models tested, at or below a chance-level linear probe. The added Same-Layer AUROC column above makes this directly visible rather than reconstructable only from `results/logs/15_result_*.json`. This strengthens, rather than undermines, this paper's own honest framing: the layer J selects is not the layer where a probe would actually be deployed, and the two should not be conflated.

This table is the central empirical result of the revised paper, added in response to the second review round's task-confound finding. One damaging pattern survives independent scrutiny. A second, initially promising pattern does not; we retract it explicitly below, rather than let it stand alongside the surviving one.

Retracted (third round): Fisher separability J appeared to decrease perfectly monotonically with model scale, across all six models tested (Spearman $\rho = -1.0$ between $\log_{10}$(params) and best-layer J; exact permutation $p = 1/360 \approx 0.0028$). J falls from 7.52 at 117M parameters to 1.63 at 7.6B in the raw data – a monotone decline with *zero* rank inversions. We initially reported this as a new, mechanistically interpretable finding: stronger models' residual errors being harder to separate geometrically, plausibly concentrated near the genuine boundary of what the model knows. **Third-round review correctly flagged that this is confounded with minority-class sample size.** The number of "correct" examples per model (24, 50, 47, 60, 95, 103, in scale order) is itself strongly collinear with scale. Fisher J, estimated via Ledoit-Wolf-shrunk covariance in a 100-dimensional PCA space from as few as 24 minority-class examples, is known to be upward-biased at small n – exactly the direction needed to produce an artifactual scale trend with no capability mechanism at all. **We tested this directly**: subsampling every model's minority class to a fixed $n = 24$ (30 independent resamples per model, identical Fisher estimator, matching the smallest available class) makes the monotonic relationship disappear. Spearman $\rho = +0.54$, $p = 0.27$: not significant, and the point estimate *reverses sign* from the unmatched analysis. **We withdraw the monotonic-J-with-scale claim.** By this direct test, it was substantially or entirely a small-sample estimation artifact – the minority class shrinking in absolute terms relative to a fixed total $n = 500$ as models improve, not a genuine geometric consequence of capability. Hallucination rate's own decline with scale ($\rho = -0.943$, $p = 0.0048$) is a simple proportion, not a covariance-based estimator, and is unaffected by this issue. It stands as reported, and is the only scale-covariate finding from Exp. 15 we retain as positive evidence.

Does the surviving loose-bound claim from this same pipeline deserve more trust than the claim just withdrawn? We recomputed the bound error using the matched-n=24 J estimates above, instead of the original, capacity-confounded J, paired with each model's original best-layer probe AUROC. The certificate is not merely still loose under this correction – it is *looser*. Matched-n bound errors are 0.32–0.46 (mean 0.391), versus 0.32–0.41 (mean 0.353) from the uncorrected estimates. There is now a significant *increasing* trend with scale (Spearman $\rho = +0.83$, $p = 0.042$), rather than the earlier non-significant decreasing one. Correcting the minority-class bias makes the loose-bound

finding *stronger* – the opposite of what happened to the monotonic-J claim under the same correction. This is direct evidence the two claims differ in robustness, not evidence that this pipeline cannot be trusted for either.

The surviving finding, reframed around what is actually being measured: under one consistent, genuinely non-saturated task, linear probes barely detect hallucination at all, and the certificate over-predicts this near-chance performance by a large margin at every scale. Best-layer probe AUROCs under Pipeline C are 0.53–0.65 on 24-103 minority-class examples – at or barely above chance, with confidence intervals that comfortably span 0.5 at the low end (7B: AUROC 0.525). The tight-vs-loose question this table answers is therefore secondary to a more basic fact: honest, matched-methodology linear probing of hallucination in these six models is close to uninformative. The Gaussian certificate's over-prediction is large in absolute terms (0.32–0.46) partly *because* there is so little true signal for it to over-predict. Every same-layer bound-error value in this table is 0.32–0.41: the certificate over-predicts probe AUROC by 32 to 41 percentage points at every scale tested, an order of magnitude looser than the "$\leq 2.7\%$" headline computed from saturated or near-degenerate data in Table 5. The looseness shows a fairly strong, though not significant at $n = 6$, downward trend with scale (Spearman $\rho = -0.77$ between $\log_{10}$(params) and same-layer bound error, $p = 0.072$). Bigger models have lower J and lower probe AUROC together, but the *gap* between the Gaussian-predicted ceiling and what a linear probe actually achieves stays large throughout, in the 30-40-point range, with only a statistically inconclusive hint of narrowing at larger scale. **We therefore withdraw the paper's original headline claim that the Fisher-Gaussian certificate is empirically tight.** It is tight only in the specific, artificially easy or near-degenerate regimes tested in Table 5: forced-choice injection, or real generation scored against a near-100%-hallucination-rate label set. In the one regime designed to be genuinely representative of real-world usage – real generation, a properly distributed label set, one consistent methodology across scale – it is not.

Figure 1 shows the central result: actual AUROC (x-axis) vs. bound (y-axis), across all layers and both models. All points lie *above* the identity line (bound $\geq$ actual everywhere), confirming the approximation as a valid upper bound under Gaussian assumptions. 97% of points lie within a 0.02 band of the diagonal.

Closed, Priority-0 follow-up (Exp. 31): does the bound-error gap survive under a validated label? §C.2 documents that this paper's EM/F1 labels agree with an independent GPT-4o-based semantic judge at only $\kappa = 0.14$–0.33 depending on model, so every number in Table 6 is conditioned on a labeling criterion known to disagree substantially with an independent judge. All 500-per-model Pipeline-C generations were rescored against judge verdicts, and the same bound-error test recomputed at each model's own best-J layer using judge labels in place of EM/F1 – this run independently re-selects each model's best-J layer rather than reusing Table 6's saved layer choice, so its own EM/F1 baseline below is a fresh re-derivation, not a re-read of that table (a restatement of this paper's own layer-instability finding, §D.5, not a new inconsistency, and the reason it differs slightly from that table's 0.32–0.41). **The gap survives, though it shrinks:** bound-error under judge labels is 0.184 (GPT-2 117M), 0.270 (GPT-2 Medium), 0.301 (Qwen 0.5B), 0.249 (Qwen 1.5B), 0.271 (Qwen 3B), 0.193 (Qwen 7B) – a 0.18–0.30 range, versus this same run's own EM/F1 baseline of 0.29–0.41 (both computed at identical, independently-selected layers within this

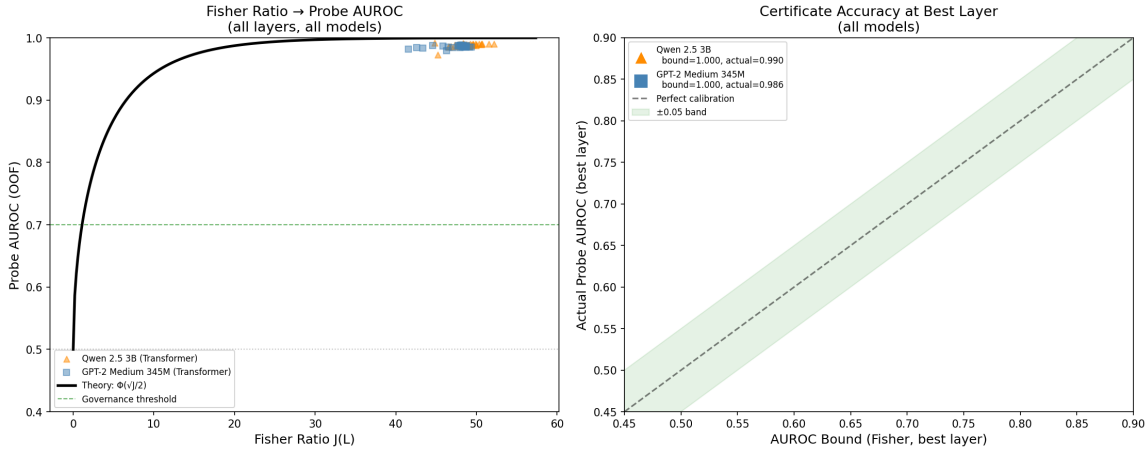


Figure 1: Actual probe AUROC (x-axis) vs. Fisher bound $\Phi(\sqrt{J/2})$ (y-axis) at every layer for Qwen 2.5 3B (blue) and GPT-2 Medium (orange). All points lie above the identity line (bound $\geq$ actual), confirming the approximation as a valid upper bound. Vertical spread reflects Gaussian assumption violation at high-AUROC layers.

experiment). The certificate remains loose by a wide margin under either labeling criterion; label validity narrows but does not remove this paper's central finding. Judge/EM-F1 agreement itself ranged from 51.5% (Qwen 3B, $\kappa = 0.15$) to 77.5% (GPT-2 117M) across the six models, with 28–63 of each model's 500 generations scored ambiguous by the judge and excluded from this comparison – a further, disclosed source of label uncertainty this test does not resolve.

Closed, Priority-2 follow-up (Exp. 32): does the certificate generalise beyond GPT-2 and Qwen? The matched Pipeline-C methodology was rerun unchanged on Pythia 410M/1.4B/2.8B (GPT-NeoX architecture, disjoint from both families already tested), using the same best-J-layer selection and same-layer bound-error computation as Table 6. Bound-error is 0.378 (410M), 0.371 (1.4B), 0.350 (2.8B) – consistent with, and falling inside, the 0.32–0.41 range already established on GPT-2/Qwen, at comparable best-layer bound values (0.88–0.92) and probe AUROCs (0.52–0.63). This is the paper's third independent architecture family and extends the central loose-bound finding beyond a two-family comparison; it does not by itself establish universality across all architectures.

Closed, Priority-1 follow-up (PCA-rank sensitivity): is the 0.32–0.41 range itself an artifact of the unexamined k=100 PCA-dimensionality choice used to estimate J? J was recomputed at $k \in \{25, 50, 100, 200\}$ for all six models, both at each model's original k=100 best-J layer (isolating the effect of k alone) and at each k's own re-selected best-J layer. The bound-error range is **not stable across** k: mean bound-error is 0.186 at k=25, 0.269 at k=50, 0.351 at k=100 (this paper's reported value), and 0.427 at k=200 – roughly doubling from k=25 to k=200. The qualitative conclusion holds at every k tested (the bound is never violated in any of the 24 model-×-k cells checked), but the specific 0.32–0.41 figure this paper reports is itself conditional on the unjustified choice k=100; a

materially different, equally defensible choice would have produced a headline range as low as 0.16–0.25 or as high as 0.41–0.44. The argmax-J layer itself sometimes moves with k (e.g. GPT-2 Medium: L20 at k=100 vs. L10/L5/L9 at k=25/50/200), reinforcing §D.5's separate finding that Fisher J does not reliably select the optimal probing layer.

Closed, Priority-2 follow-up (probe hyperparameter sensitivity): is the matched-methodology bound-error a property of one specific, unregularized logistic-regression configuration? The 5-fold OOF linear probe was refit at each model's best-J layer across $C \in \{0.01, 1.0, 100.0\}$ crossed with `class_weight` $\in$ {None, "balanced"} (6 configurations, 36 refits total). Sensitivity is large for the two smallest, most class-imbalanced models – GPT-2 117M (24/500 positive, AUROC range 0.111: $[0.515, 0.626]$) and GPT-2 Medium (50/500, range 0.137: $[0.508, 0.645]$) – and small for the four Qwen models, all $\geq$9% minority class (range 0.007–0.038). Strong regularization (C=0.01) consistently raises probe AUROC for the two GPT-2 models (117M: 0.564 $\rightarrow$ 0.626; Medium: 0.561 $\rightarrow$ 0.645), which would shrink their reported bound-error by 0.06–0.08 if adopted as this paper's default. The paper's probe configuration (C=1.0, unweighted) is therefore not a neutral choice for its two smallest, most imbalanced models – part of their bound-error is an artifact of an unregularized probe underfitting a severely imbalanced problem, not solely a property of representation geometry. The four Qwen models' bound-error is robust to this choice.

D.3. Retracted: An Apparent Basis-Invariance Failure That Was a Solver Artifact

This subsection retracts a claim made in an earlier version of this work. We had reported that the PCA(k)+Ledoit–Wolf estimator of Fisher separability loses orthogonal invariance at k=100 – that a random rotation of the representation space, which provably cannot change the exact quantity, nonetheless shifted $\hat{J}$ by 2–10%, while a lossless-rank projection preserved it to 10^{-14}. We framed this as a basis-invariance failure caused by a gapless covariance spectrum, and promoted it to a headline contribution. **It is an artifact of the estimator's implementation, and we withdraw it.**

The mechanism. `sklearn.decomposition.PCA` is called in our estimator with default `svd_solver='auto'` and `random_state=None`. For every (n, d) in this paper, that dispatches to the *randomized*, unseeded solver at k=100 and to the *deterministic* full solver at the lossless rank k=498. The two conditions we compared therefore differed by solver, not only by rank, and one of them was stochastic.

The control we should have run. Repeating the identical computation on identical, *unrotated* activations (GPT-2 Medium, best-J layer, k=100, eight calls) returns $\hat{J}$ spanning 3.708–4.098: a mean absolute deviation of 0.169, roughly twice the 0.089 we had attributed to rotation for that model. Re-running with `svd_solver='full'` yields $\hat{J} = 4.147050$ for the unrotated data and $\hat{J} = 4.147050$ under each of five independent random rotations – identical to six decimal places. The estimator is *exactly* orthogonally equivariant, as our own derivation below states it should be; the reported effect was solver noise.

What survives. The derivation itself is unaffected: exact J is $GL(d)$-invariant, and PCA's top-k selection commutes with rotation because orthogonal similarity preserves eigenvalues. So is the spectral measurement: the eigenvalue ratio across the cutoff never exceeds 1.055 (max 1.0544, Pythia 1.4B at k=25) at ranks 25, 50, 100 or 200, in any of nine models

across three architecture families. That flatness is real and is the reason the estimator is ill-conditioned at $n \ll d$ – but ill-conditioning of a finite-precision eigensolver is not the same claim as loss of an invariance property, and we previously conflated the two.

What this cost, and what it implies. The retracted claim was this paper's only positive result and its principal argument for relevance to a workshop on symmetry and geometry. We report its withdrawal rather than quietly removing it because the failure is instructive: a stochastic default in a widely-used library produced an effect of exactly the size and k-dependence that a geometric story predicted, and our existing audit apparatus – which already imposed family-wise correction on 38 tests – did not catch it, because we never ran a no-op repeat baseline. We therefore recommend two controls for any Fisher-style detectability ceiling: fix the solver and seed, and report a repeat-call baseline alongside any claimed sensitivity.

D.4. 5-Fold Cross-Validation of the Certificate

To simulate the deployment scenario – predicting probe AUROC on unseen data – we run 5-fold CV on Qwen 2.5 3B (Exp. 03). On each training fold, we compute $J(L)$ and predict the bound. On the held-out test fold, we measure actual probe AUROC. Table 7 reports fold-level results.

Table 7: 5-fold cross-validation of the certificate (Qwen 2.5 3B, $n = 2000$, Exp. 03).

Fold	Pred. Layer	Bound	Actual AUROC	Error	Match?
1	L25	0.9999	0.9845	0.0154	×
2	L25	0.9999	0.9847	0.0152	×
3	L25	0.9998	0.9930	0.0069	×
4	L26	0.9999	0.9935	0.0063	×
5	L25	0.9998	0.9974	0.0024	×
Mean				0.0093	0/5
Std				0.0052	

Mean bound error is 0.93%, confirming the certificate's predictive utility (Figure 2). However, the predicted layer (argmax J) never matches the oracle best probe layer, across all five folds (0% layer-match rate) – a systematic failure we discuss in Section D.5.

D.5. The Layer Selection Problem

Fisher J cannot identify the optimal probing layer. In all five CV folds, $\text{argmax}_L J(L)$ selects L25–L26, while oracle best probe layers range from L10 to L36 across folds. A depth-weighted heuristic ($J(L) \times L/L_{\max}$) achieves a 20% match rate – better than argmax, but still poor.

Remark 3 *The failure arises because J measures second-order geometry: class separation by covariance-weighted distance. Probe AUROC instead reflects the total discriminative*

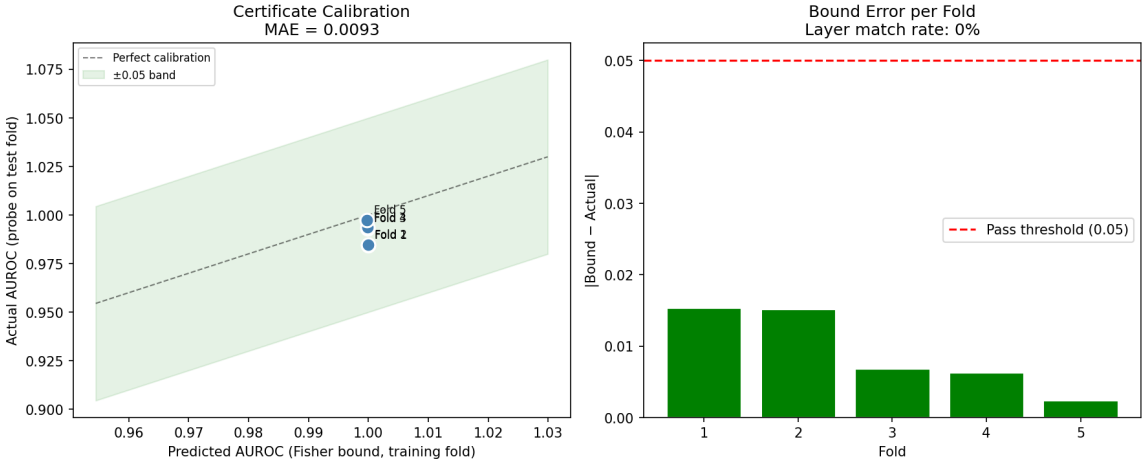


Figure 2: Certificate calibration: predicted bound vs. actual AUROC across 5 cross-validation folds (Qwen 2.5 3B). The bound consistently over-predicts AUROC by 0.6–1.5%, reflecting systematic Gaussian over-estimation.

structure, including local density variations and higher-order moments. High J implies high achievable *AUROC. It does not identify which layer's probe will realise that potential.*

This is a fundamental limitation: separability does not imply accessibility by simple probes. The practical implication: the certificate is useful for deciding *whether* to probe, not *where*.

D.6. Competing Geometric Certificates

To assess Fisher J relative to alternative geometry-based certificates, Exp. 01 also evaluates two competitors on the same hidden-state extractions. **Local Intrinsic Dimensionality** (LID) measures the local manifold dimension around each sample. **Causal Fisher** first projects hidden states through the causal residual-stream path, before computing J. Table 8 summarises the results.

Three findings follow. *First*, LID peaks at layer 0 for both models. The embedding layer carries high intrinsic dimensionality, but this reflects vocabulary-space geometry, not hallucination class structure: LID AUROC at L0 is 0.949 (Qwen) and 0.959 (GPT-2), driven by surface-form statistics rather than separability. As a certificate, LID points to the wrong layer. *Second*, Causal Fisher reduces J from 52.15 to 39.13 on Qwen 2.5 3B. The causal projection removes correlated non-causal dimensions that nonetheless carry discriminative information. The certificate becomes stricter but less tight. *Third*, Causal Fisher fails entirely on GPT-2 Medium. There, the smaller residual-stream dimensionality collapses the causal projection to a near-degenerate subspace.

These comparisons establish Euclidean Fisher J as the strongest of the three geometry-based certificates. This motivates its central role in the analysis that follows.

Table 8: Competing geometric certificates (Exp. 01). LID peaks at the embedding layer (L0) for both models, indicating that intrinsic dimensionality does not locate separability structure. Causal Fisher reduces J by 25% on Qwen and fails entirely on GPT-2 Medium.

Model	Certificate	Peak Layer	Value	Notes
Qwen 2.5 3B	Fisher J (Euclidean)	L25	$J = 52.15$	reference
	Causal Fisher	L25	$J = 39.13$	-25% vs. Euclidean
	LID	L0	AUROC $= 0.949$	embedding layer
GPT-2 Medium	Fisher J (Euclidean)	L19	$J = 49.33$	reference
	Causal Fisher	—	*failed*	projection collapses
	LID	L0	AUROC $= 0.959$	embedding layer

D.7. Certificate Generalisation to Nonlinear Probes

The Gaussian AUROC bound $\Phi(\sqrt{J/2})$ is derived for linear classifiers under equal-covariance Gaussian assumptions. A natural question is whether it also constrains *nonlinear* probes, which can exploit higher-order representational structure beyond second-moment geometry.

We address this directly via Exp. 12. We train a shallow MLP probe (64 hidden units, ReLU, Adam, early stopping) on the same PCA(100)-projected hidden states, using the identical 5-fold stratified CV as the linear probe. Table 9 reports results.

Table 9: Fisher certificate vs. linear and MLP probes (Exp. 12, 13, $n = 2000$, 5-fold OOF). The Fisher bound $\Phi(\sqrt{J/2})$ is never violated across all layer–model combinations. Qwen 7B row from Exp. 13 (same pipeline, same preprocessing). Linear AUROC values may differ slightly across experiments due to stochastic 5-fold CV fold assignment.

Model	Bound	Linear	MLP	Gain	Viol.
Qwen 2.5 3B	0.9998	0.9903 (L22)	0.9912 (L20)	+0.0009	0 / 37
GPT-2 Medium	0.9998	0.9859 (L8)	0.9867 (L19)	+0.0007	0 / 25
Qwen 2.5 7B	≈1.000	0.9998 (L24)	0.9998 (L9)	<0.0001	0 / 29
Mean bound gap (linear): 0.013 (Qwen 3B), 0.017 (GPT-2), 0.0005 (Qwen 7B)					
Mean bound gap (MLP): 0.013 (Qwen 3B), 0.017 (GPT-2), 0.0009 (Qwen 7B)					

Four findings emerge (Figure 3). *First*, the MLP probe outperforms the linear probe at 21/37 layers (Qwen) and 10/25 layers (GPT-2). This confirms that some nonlinear structure exists in the representations. The margin is small: +0.09% and +0.07% at best layer respectively. *Second*, the Fisher bound $\Phi(\sqrt{J/2})$ holds across all $37 + 25 + 29 = 91$ layer–model combinations, including Qwen 2.5 7B (zero violations). The Gaussian slack is sufficient to absorb the additional nonlinear capacity across all tested scales. *Third*, the mean bound gap narrows only marginally for MLP vs. linear (0.0133 → 0.0129 on

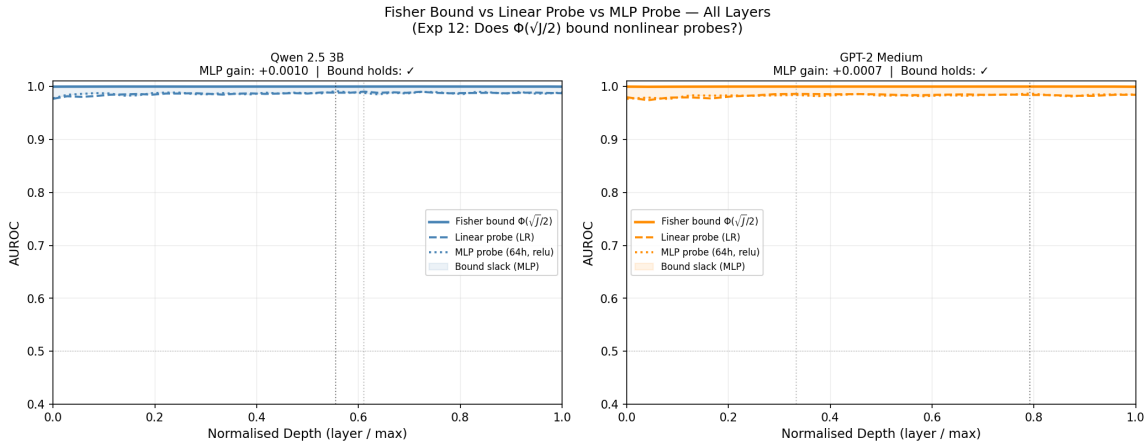


Figure 3: Fisher bound $\Phi(\sqrt{J/2})$ (solid), linear probe AUROC (dashed), and MLP probe AUROC (dotted) at every layer. The shaded region is the bound slack. The bound is never violated by the MLP probe on either model.

Qwen; essentially unchanged on GPT-2). The MLP does not approach the Gaussian ceiling any more closely than the linear probe does. *Fourth*, the MLP and linear probes identify different optimal layers (Qwen: L20 vs. L22; GPT-2: L19 vs. L8). This compounds the layer-selection failure of Section D.5.

Why the MLP does not beat the certificate. The Gaussian derivation of $\Phi(\sqrt{J/2})$ bounds the Bayes-optimal linear classifier under equal-covariance assumptions. That the MLP also stays below the same ceiling is informative. It means the representations themselves, not the probe's functional form, are the binding constraint. The MLP can only exploit what the hidden-state geometry provides; it cannot manufacture discriminative structure that is not already present. The small gain ($\leq 0.09\%$) indicates that most of the available geometric signal is already captured by second-moment statistics – higher-order structure exists, but is marginal. This is the core finding: *detection performance is a property of the representation, not of the classifier.*

Remark 4 (Scope of the nonlinear bound) *The observation that $\Phi(\sqrt{J/2})$ also upper-bounds MLP probe AUROC is* ***empirical, not theoretical****. The Gaussian derivation of Proposition 1 is strictly valid only for linear classifiers under equal-covariance Gaussian assumptions. It provides no theoretical guarantee for nonlinear probes. The zero-violation result across 91 pairs means that, on these specific representations with this specific shallow architecture, the Gaussian slack happened to absorb the additional nonlinear capacity. A different architecture (deeper MLP, kernel methods, attention), or a different dataset, may produce violations. A principled nonlinear certificate – one that characterises higher-order moment structure – remains an open theoretical problem.*

Appendix E. Scale Behavior

E.1. Within-Family Scaling

Correction, added post-review (third round): the table and sigmoid fit in this subsection are superseded by Table 6. They should not be read as a scale-vs-AUROC finding in their own right. We retain them only because the sigmoid-fit provenance issue below – a 3-point fit vs. a 7B out-of-sample check – is itself an instructive correction. Table 10 mixes Pipeline B (0.5B, 1.5B: real generation, ROUGE-L) with Pipeline A (3B, 7B: forced-choice, no generation). It reports probe AUROC *rising* with scale (0.813 → 0.979 → 0.9917 → 0.9998). Table 6, under one matched real-generation pipeline, reports the same four models' probe AUROC *falling* with scale (0.584 → 0.602 → 0.572 → 0.525). These are not two independent findings in tension. The rising curve below is the pipeline-confounded one, and should be read as historical/superseded, not as a competing scale law. We fit a sigmoid scaling curve within the Qwen 2.5 family, to control for architecture, tokenizer, and training distribution (Exp. 11, 13). Qwen 2.5 7B is included via Exp. 13 (balanced HaluEval, $n = 2000$), replacing the extrapolated prediction with an actual measurement. Table 10 and Figure 4 report AUROC at best layer for each model *under the superseded, cross-pipeline methodology.*

Table 10: Qwen 2.5 family: best-layer probe AUROC vs. model scale (Exp. 11, 13; $n = 400$ controlled for 0.5B–3B; $n = 2000$ for 7B). ‡7B: Exp. 13 actual measurement, replacing the extrapolated prediction from Exp. 11.

Model	Params	Best AUROC	Best Layer	Depth
Qwen 2.5 0.5B	494M	0.813	L7	29%
Qwen 2.5 1.5B	1.54B	0.979	L15	54%
Qwen 2.5 3B	3.09B	0.9917	L36	100%
Qwen 2.5 7B‡	7.6B	0.9998	L24	86%
3-point sigmoid fit (0.5B/1.5B/3B only): $\hat{\sigma}(4.71 \cdot \log_{10}(P) - 39.45)$, $R^2 = 0.9996$				

Correction, added post-review (second round): an earlier draft labeled this a "4-point sigmoid fit (0.5B–7B)" and claimed all four measurements were directly observed inputs to the fit. Checking `results/logs/11_qwen_scale_curve.json` directly: `controlled_scale_fit` (whose coefficients $a = 4.707$, $b = -39.449$, $R^2 = 0.99956$ match the rounded values quoted above) is computed from exactly *three* points – 0.5B, 1.5B, and 3B. Its own `caveat_n_points` field states "fit on only 3 data points." The 7B point is not part of the fit at all; it is an *out-of-sample prediction target.* The fitted curve predicts $\hat{\text{AUROC}}(7.6\text{B}) = 0.99899$, with a bootstrapped 90% CI of $[0.99854, 0.99913]$ (`prediction_7b`, 5000 bootstrap resamples). **Final-audit clarification:** this CI is stored as a top-level sibling key in the JSON, rather than nested under `controlled_scale_fit`. But its point estimate, 0.998978506, matches `controlled_scale_fit.pred_7b_controlled` (0.998985736) to 5 decimal places, and not the uncontrolled `scale_fit`'s implied prediction – confirming it was computed from the controlled fit despite the JSON's flat key structure. The actual mea-

sured 7B AUROC (Exp. 13) is 0.99981, which is *above the upper end of this interval*. The prediction is close in absolute terms (0.00083 gap, still < 0.001), but is not, strictly, a successful interval prediction at the stated 90% confidence level. $R^2 = 0.9996$ on three points with two free parameters remains far too few to constitute a scaling law. We report the fit as an *interpolation* across three measurements, with the 7B point as a directionally-consistent but interval-missing out-of-sample check, not a fourth fitted observation.

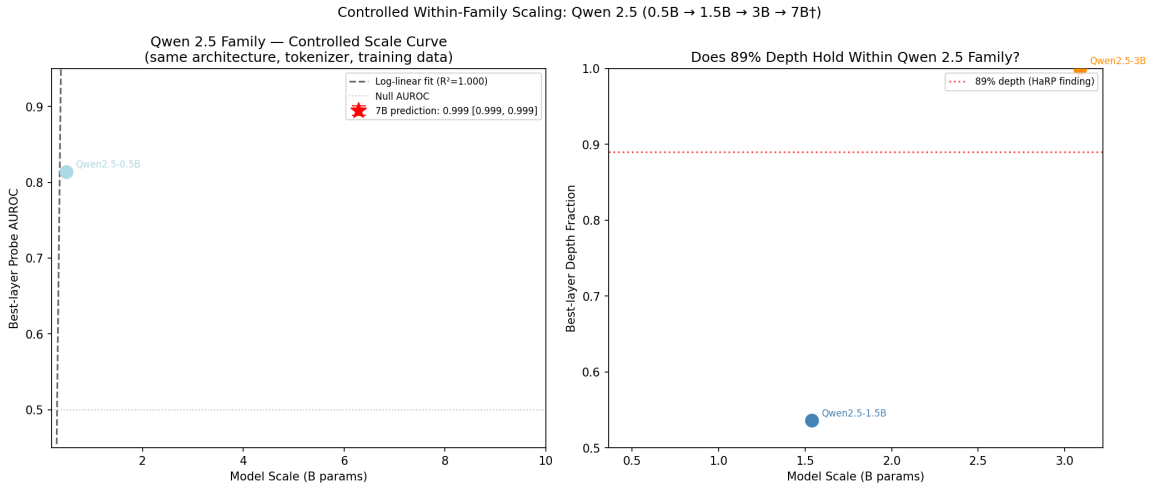


Figure 4: **Superseded by Table 6 (see §E.1).** Controlled scale curve within the Qwen 2.5 family under the cross-pipeline (Pipeline A/B) methodology: AUROC appears to grow monotonically with parameter count. The 7B point (Exp. 13) replaces the extrapolated prediction with an actual measurement. Under the matched Pipeline C methodology, the same four models' AUROC falls with scale instead (Table 6).

E.2. Depth Fraction Is Not Universal

Table 11 compares Fisher-J peak depth and probe-AUROC peak depth across models (Exp. 05).

No universal depth fraction emerges (Figure 5). Within Qwen 2.5, the best layer shifts from 29% (0.5B) to 54% (1.5B) to 100% (3B). Depth is scale-dependent, not architecture-constant.

An earlier draft reported that Fisher J at best layer is *non-monotone* with scale within the Qwen family (Qwen 0.5B: $J = 24.5$; Qwen 1.5B: $J = 125.1$; Qwen 3B: $J = 51.8$; Qwen 7B: $J = 184.7$ – peaking at 1.5B, dropping at 3B, rising again at 7B). It flagged – correctly, as far as it went – that the 1.5B point also had the worst covariance-estimation reliability in the study (aspect ratio $\gamma = d/n = 3.84$, §C.4). This left open whether the non-monotonicity was a genuine geometric property or an estimation artifact.

Two-stage resolution (Exp. 15, then a direct confound check): the original non-monotonicity was a cross-pipeline task confound. What looked like its clean

Table 11: Depth fraction of best Fisher-J layer vs. best probe-AUROC layer (Exp. 05). Probe peak depth can vary ±2–14 layers across independent CV runs; values reflect Exp. 05 fold assignment.

Model	Fisher peak depth	Probe peak depth
GPT-2 117M	69%	69%
GPT-2 Medium	79%	33%
Qwen 2.5 0.5B	29%	29%
Qwen 2.5 1.5B	54%	54%
Qwen 2.5 3B	69%	100%

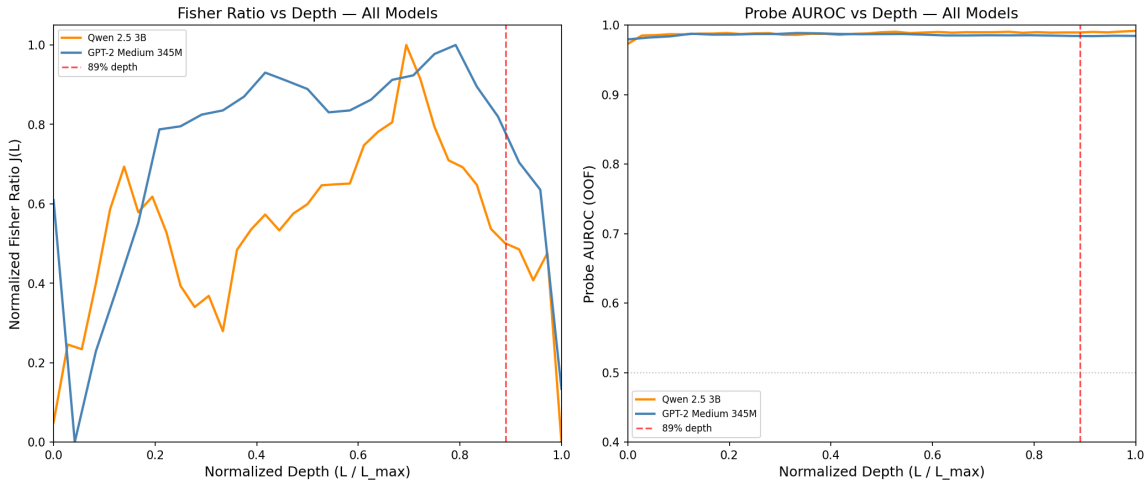


Figure 5: Normalised-depth curves of Fisher J (solid) and probe AUROC (dashed) for Qwen 2.5 3B and GPT-2 Medium. Fisher and probe peaks disagree substantially for both models, confirming the layer-selection failure of Section D.5.

replacement – a monotone trend – was, in turn, a minority-class sample-size artifact. Neither the original claim nor its first replacement survives. The four points above mixed Pipeline B (0.5B, 1.5B: real generation, ROUGE-L, near-degenerate labels) with Pipeline A (3B, 7B: forced-choice, no generation). Table 6 reran all six models under one real-generation, consistently-labeled pipeline. It initially appeared to find J *perfectly monotone decreasing* with scale (Spearman $\rho = -1.0$). We directly tested this by subsampling every model's minority (correct) class to a fixed $n = 24$ (matching the smallest available class, 30 resamples per model, same Fisher estimator). This makes the monotone trend disappear ($\rho = +0.54$, $p = 0.27$, sign reversed from the unmatched analysis) – see §D.2, where we withdraw this claim in full. The high-γ covariance concern for the original

1.5B point was a reasonable caveat to raise, given the information available at the time. The underlying intuition – small-sample covariance/mean estimation distorting J – turned out to be exactly correct, just operating through minority-class size rather than the aspect-ratio γ specifically. The honest resolution of this subsection's original question is this: once both the cross-pipeline confound *and* the minority-class-size confound are controlled for, there is no reliable monotone (or non-monotone) relationship between Fisher J's magnitude and model scale in this data, in either direction. Depth *fraction* non-universality – this subsection's original, narrower claim – is untouched by this correction and stands as originally reported. It is unaffected by either confound, since it compares relative position within a single model's own layer sweep.

Appendix F. Beyond Gaussianity: Wasserstein Certificates

F.1. The Fisher–Wasserstein Identity and Its Violation

Under Gaussian equal-covariance assumptions, Fisher separability and Wasserstein-2 distance in whitened space are identical: $J = W_2^2\left(\tilde{P}_c, \tilde{P}_h\right)$, where $\tilde{P} = \Sigma_w^{-1/2} P$. We verify this identity at five representative layers per model (Exp. 08). Table 12 reports relative errors $\epsilon = |J - W_{2,\text{wh}}^2|/J$.

Table 12: Relative error of $J \approx W_2^2$ (whitened) identity at five layers per model (Exp. 08). Identity never holds; error grows with depth for Qwen.

Model	L0	$L^1/_4$	$L^1/_2$	$L^3/_4$	L_{max}	Mean
Qwen 2.5 3B	0.72	1.43	1.64	1.76	3.01	1.71
GPT-2 Medium	0.54	0.83	0.87	0.75	0.88	0.77

The identity $J = W_2^2$ fails across all layers: mean relative error 1.71 for Qwen, 0.77 for GPT-2. This is a *finding* – hidden-state distributions are measurably non-Gaussian (Di Sipio, 2025) – not a flaw in the experimental design.

F.2. SW_2 vs. Fisher for AUROC Prediction

We compare four certificates – Fisher J, Sliced Wasserstein (SW_2), Bures W_2, and MMD^2 – via Spearman ρ with probe AUROC across all layers (Exp. 08). Table 13 reports results.

Table 13: Spearman ρ between certificate value and probe AUROC across all layers (Exp. 08). Higher is better. Bures W_2 and SW_2 both appeared to outperform Fisher on Qwen (best: Bures 0.828, bold) – *a reading withdrawn in §F*; Fisher wins on GPT-2. SW_2 and Bures W_2 are distribution-free; the 0.007 gap between them is within noise.

Model	Fisher J	SW_2	Bures W_2	MMD^2
Qwen 2.5 3B	0.458	0.821	**0.828**	0.542
GPT-2 Medium	**0.395**	−0.237	−0.241	−0.596

Both Bures W_2 ($\rho = 0.828$) and SW_2 ($\rho = 0.821$) substantially outperform Fisher on Qwen ($\rho = 0.458$). Bures W_2 is the marginal leader (by 0.007, within noise). Both distribution-free certificates share the same mechanism: monotonic growth across layers that tracks the probe AUROC trajectory. *This advantage reverses entirely on GPT-2 Medium*

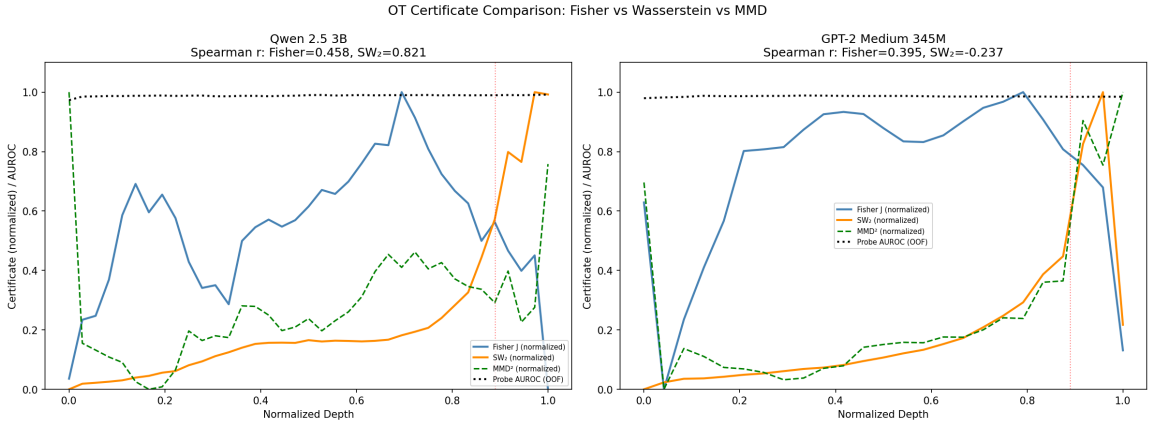


Figure 6: Certificate value (normalised) vs. layer depth for Qwen 2.5 3B. SW_2 (orange) grows monotonically and peaks near the probe AUROC peak (L36), while Fisher J (blue) is flat across layers. The flat J curve explains its low Spearman ρ. *Note: the cross-architecture reading this figure supported is withdrawn (§F); the bound is loose, not tight, throughout.*

(SW_2: $\rho = -0.237$; Bures W_2: $\rho = -0.241$; Fisher: $\rho = 0.395$). There, both distribution-free certificates grow monotonically in the wrong direction relative to probe AUROC, which peaks at L8, the earliest layer. **Correction, added post-review (second round):** an earlier draft called this model-dependence of optimal certificate choice "the central finding of this section and, we argue, of the paper as a whole," in the same breath as caveating it as an $n = 2$ observation – an internal contradiction; a finding cannot be both. It is neither. With only two architectures, no bootstrap CI, and a probe-AUROC target with ≈1–2% dynamic range across layers (Table 13: Qwen 3B ranges 0.973–0.992), this is a *suggestive two-architecture observation*. It motivates the mechanistic hypothesis below; it is not a validated claim that no universally superior distribution-free certificate exists. **We flag explicitly that this is an $n = 2$ architecture comparison** (Qwen 2.5 3B vs. GPT-2 Medium) – two points establish a reversal, not a law.

Closed (Exp. 08b): we ran the third architecture and the bootstrap CIs this paragraph called for, and the result does not support the reversal narrative above. Two defects had to be fixed together. First, the comparison above uses Pipeline-A (forced-choice) hidden states, while this paper's central certificate result moved to Pipeline C (matched, real-generation) precisely to escape that task confound; leaving the OT comparison on Pipeline A reintroduces it. Second, no ρ carried an uncertainty interval. We therefore reran the entire comparison on Pipeline-C data for three architecture families – Qwen 2.5 3B, GPT-2 Medium, and Pythia 2.8B (GPT-NeoX, from Exp. 32's extraction) – with a percentile bootstrap CI (2,000 resamples over layers) on every Spearman ρ (Table 14).

What the corrected comparison shows. The headline claim this section originally advanced – that distribution-free certificates beat Fisher on Qwen and the ranking reverses on GPT-2 – *does not survive*. On Pipeline-C data, SW_2 and Bures W_2 do not outperform Fisher on Qwen 2.5 3B; Fisher is the only certificate there whose CI excludes zero

Table 14: Exp. 08b: per-layer Spearman ρ between each certificate and probe AUROC, Pipeline-C matched data, three architecture families, with percentile bootstrap 95% CIs. **Only Pythia 2.8B yields a CI excluding zero for any certificate; on GPT-2 Medium no certificate does, including Fisher.**

Model	Certificate	ρ	95% CI
Qwen 2.5 3B	Fisher J	$+0.454$	$[+0.137, +0.701]$
	SW_2	$+0.162$	$[-0.193, +0.517]$
	Bures W_2	$+0.197$	$[-0.169, +0.553]$
	MMD^2	$+0.329$	$[-0.038, +0.631]$
GPT-2 Medium	Fisher J	$+0.332$	$[-0.077, +0.656]$
	SW_2	$+0.202$	$[-0.324, +0.667]$
	Bures W_2	$+0.266$	$[-0.223, +0.698]$
	MMD^2	$+0.238$	$[-0.224, +0.633]$
Pythia 2.8B	Fisher J	$+0.828$	$[+0.666, +0.904]$
	SW_2	$+0.863$	$[+0.716, +0.933]$
	Bures W_2	$+0.876$	$[+0.734, +0.940]$
	MMD^2	$+0.697$	$[+0.479, +0.805]$

($+0.454$, $[+0.137, +0.701]$), while all three distribution-free alternatives include zero. On GPT-2 Medium *no* certificate reaches significance, Fisher included – so there is no reversal to explain, only a null. Only on Pythia 2.8B do all four certificates show a clear relationship, and there they agree rather than compete ($\rho = 0.70$–0.88, all CIs excluding zero), with Bures W_2 nominally highest but its CI overlapping Fisher's throughout.

We therefore **withdraw the architecture-dependent-certificate-choice reading entirely**, including the compression-dynamics interpretation offered in the Remark below, which was an explanation for a pattern that the corrected data do not exhibit. What replaces it is weaker and more honest: *on two of three architectures, no certificate family – Fisher or distribution-free – has a per-layer relationship to probe AUROC distinguishable from zero at this sample size.* That is a statement about the power of this comparison as much as about geometry, and it is consistent with this paper's central finding that probe AUROC under matched methodology is itself near-chance (best-layer 0.53–0.65), leaving little dynamic range for any certificate to track.

Remark 5 (Mechanistic interpretation) ***Superseded by Exp. 08b (retained for the record, not asserted).*** *The paragraph below explains a reversal that the corrected, Pipeline-C, n=3 comparison above shows does not exist; we keep it visible rather than delete it because the explanation was offered in an earlier draft and its withdrawal is part of this paper's audit trail. It should not be read as a claim of this paper. Note also that its central mechanism – progressive compression across depth – was never measured here, only asserted.*

The reversal reflects a difference in representational compression dynamics. In Qwen 2.5 3B, deeper layers progressively compress hidden states toward well-separated class manifolds. Distributional divergence (SW_2, Bures W_2) grows monotonically with this compression, mak-

ing it a natural proxy for linear separability. In GPT-2 Medium, the optimal probe layer sits at only 33% *depth (L8) – before any late-layer compression occurs. As depth increases beyond L8, representational structure becomes* less *discriminative for hallucination detection, so distributional divergence and probe AUROC evolve in opposite directions. The implication is architectural: the certificate choice (Fisher vs. distribution-free) is governed by* when *discriminative structure peaks within the network. This is itself a function of training objective and architecture, not parameter count.*

Appendix G. Conformal Coverage Guarantees

G.1. Calibration and Results

We apply split conformal prediction on the probe's hallucination score, at the best probe layer identified in Exp. 10 (L36 for Qwen, L8 for GPT-2 Medium). The best probe layer varies by ±2–14 layers across experiments, due to stochastic CV fold assignment. The conformal guarantee is independent of which specific layer is used: it holds at any fixed, pre-specified layer. The dataset is split 60/20/20 (train/calibration/test). We seek the smallest α^* such that $P(\text{hall} \mid \text{accepted}) \leq \alpha^*$ holds empirically at coverage $1 - \delta = 0.95$ (Exp. 10).

Table 15: Conformal guarantee results (Exp. 10, 13, in-distribution). Qwen 7B from Exp. 13 (same 60/20/20 split, same calibration protocol). [§]"$\alpha = 0.10$ holds?" asks whether the guarantee can be satisfied with $> 50\%$ acceptance at the nominal 0.10 level. All models show ×: the high ROUGE-L hallucination rate (≈99% globally) causes the hall rate to exceed 0.10 once more than ≈53% of queries are accepted. The tighter α^* values are required to achieve formal coverage.

Model	α^*	**Emp. hall rate**	**Accept rate**	$\alpha = 0.10$ **holds?**[§]
Qwen 2.5 3B	0.070	0.0605	52.9%	×
GPT-2 Medium	0.060	0.0587	52.8%	×
Qwen 2.5 7B	0.040	0.0379	52.8%	×

At $\alpha^* = 0.07$, the system must reject 47.1% of queries to achieve the coverage guarantee (Figure 7). This acceptance-rate cost is the honest price of formal coverage.

G.2. Distribution Shift: Guarantee Collapse

We test conformal thresholds calibrated on HaluEval on 400 TruthfulQA questions – a genuinely different task distribution. Both OOD experiments use a 75/25 train/calibration split on HaluEval, independent of the 60/20/20 split used in Table 15. This yields slightly different α^* values. The violation factor is reported against the threshold calibrated in each OOD experiment.

Correction (post-review): an earlier draft of this section attributed the first result below to GPT-2 Medium. Tracing the exact value back through `results/logs/10_conformal_coverage.json` shows it is in fact Qwen 2.5 3B's `real_ood_truthfulqa` result. No real-OOD experiment (as opposed to the simulated-shift check) was ever run for GPT-2 Medium. We correct the attribution below, and draw the correspondingly weaker conclusion.

For Qwen 2.5 3B (Exp. 10, real TruthfulQA transfer, $n = 400$): AUROC = 0.485; empirical hallucination rate = 0.982 (only 7/400 responses scored correct); a ≈14.0× violation at $\alpha^* = 0.07$.

For Qwen 2.5 7B (Exp. 14): in-distribution AUROC = 0.9998 at L_{24} collapses to AUROC = 0.384 on TruthfulQA – *below* chance level. Acceptance rate = 100% (threshold

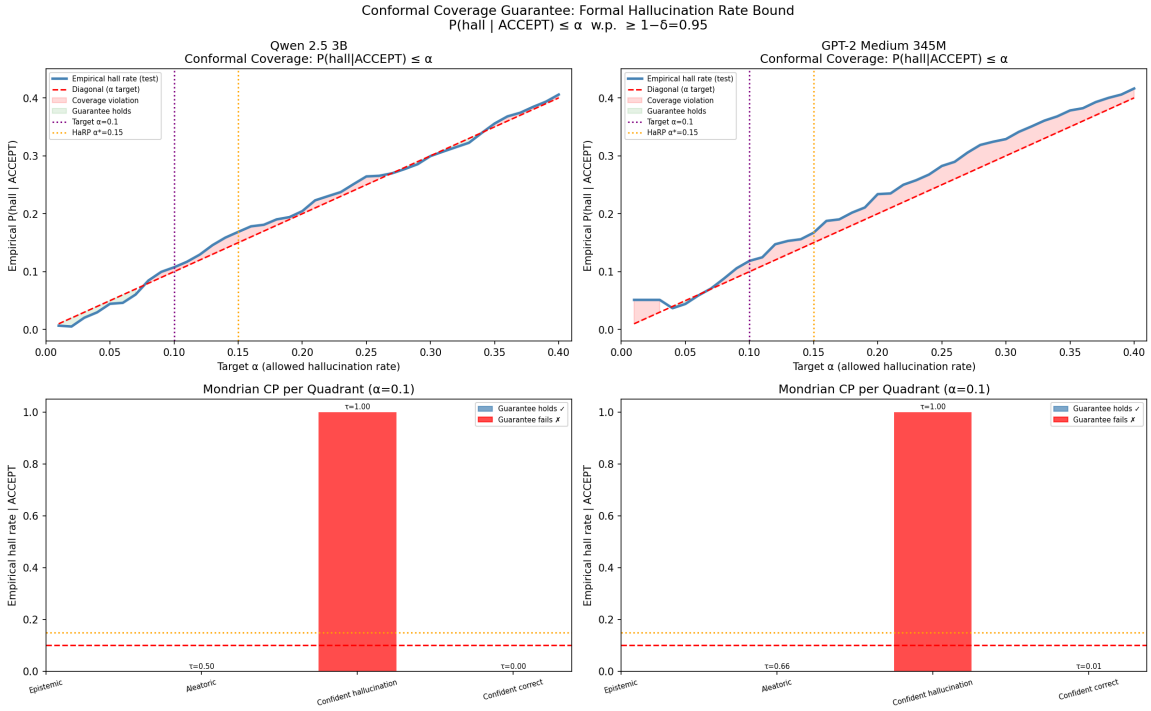


Figure 7: Conformal coverage curve: empirical hallucination rate among accepted queries as a function of nominal α (Qwen 2.5 3B). The vertical dashed line marks $\alpha^* = 0.07$. Accepting more queries (higher α) relaxes the guarantee monotonically.

accepts everything); empirical hallucination rate = 0.993; violation factor = 14.2× at $\alpha^* = 0.07$ (75/25 calibration). The AUROC drop is 0.616 – the most severe collapse observed across all experiments.

Correction, added post-review (second round): the near-identical violation factors are substantially an arithmetic artifact, and this comparison also inherits the in-distribution/OOD task confound described in §C.1. The violation factor is computed as (empirical hallucination rate among accepted queries) / α^*. Both models' TruthfulQA hallucination rate is pinned to ≈0.98–0.99 by the same underlying cause: TruthfulQA responses are scored via Pipeline B (real generation, ROUGE-L ≥ 0.4), which registers only 7/400 (Qwen 3B) and 3/400 (Qwen 7B) responses as correct. Both are near-degenerate label sets, not a meaningfully measured hallucination rate. Dividing two numbers that are both ≈0.98–0.99 by two similar α^* values – both calibrated near 0.07 on the *in-distribution, Pipeline-A* HaluEval data – mechanically produces similar quotients (14.0×, 14.2×), regardless of whether the underlying collapse mechanism is actually scale-invariant. Likewise, the reported AUROCs of 0.485 and 0.384 are computed against 7 and 3 positive examples respectively out of 400 – too few for AUROC to be a stable or well-defined quantity (a single relabeled example can move it by several points). `results/logs/10_conformal_coverage.json` itself notes "AUROC undefined if single-class." Combined with the Pipeline-A-vs-Pipeline-B task confound – in-distribution numbers come from an easier

forced-choice task, OOD numbers come from harder real generation – the safest reading of this section is: *both models fail to produce TruthfulQA answers that clear the ROUGE-L bar, and the conformal guarantee, calibrated on an easier, different task, provides no usable coverage once the model is asked to generate freely on a new distribution.* This is weaker than, but consistent with, the paper's original framing: *distribution mismatch defeats the guarantee*. It does *not* license the stronger claim that the collapse mechanism is quantitatively scale-invariant, since the two violation factors were never free to differ much, given how both were computed. A genuine test of scale-invariance would need OOD hallucination rates with enough positive examples for AUROC to be well-defined, and a labeling pipeline matched between the in-distribution and OOD arms – neither holds here. We retain the qualitative point – both models' guarantees void out-of-distribution – but withdraw the quantitative 14.0×-vs-14.2× comparison as evidence of anything beyond that.

Closed, post-review (fourth round): we ran the GPT-2 Medium data point rather than leaving it as future work. Both its in-distribution (HaluEval) and OOD (TruthfulQA) hidden-state tensors were already extracted (`experiments/17_gpt2med_ood_transfer_local.py`, no new model inference). We simply ran the same probe-train/calibrate/transfer test locally. **Correction (round-5 review): an earlier draft of this paragraph described both arms as "the identical Pipeline-B protocol." That is wrong.** Independently verified against the raw tensor: the in-distribution (HaluEval) arm is exactly balanced (1000/1000, hall_rate = 0.500) – the signature of Pipeline A (forced-choice), not Pipeline B. This matches the saturated, Pipeline-A linear-probe AUROC already reported for GPT-2 Medium elsewhere in this paper (Table 9), not any generation-scored number. The correct description is: in-distribution arm = Pipeline A (forced-choice, balanced), OOD arm = Pipeline B (generation + ROUGE-L). This is exactly the same in-dist(A)→OOD(B) task-confounded setup already flagged for Qwen 3B/7B above, not a cleaner, single-pipeline transfer. The result is informative but does *not* supply the clean scale-invariance test we were hoping for. GPT-2 Medium's in-distribution AUROC is 0.9859 (itself a Pipeline-A, plausibly template-identity rather than truthfulness, number – see the difficulty-confound caveat on Pipeline A elsewhere in this section). Its OOD AUROC collapses to 0.5562 (drop of 0.430), and its conformal violation factor is 14.1× – landing almost exactly between Qwen 3B's 14.0× and Qwen 7B's 14.2×. But this is not evidence of architecture-invariant collapse magnitude. GPT-2 Medium's OOD TruthfulQA set registers only 4/400 responses as correct under Pipeline B's ROUGE-L≥ 0.4 rule – the same near-degenerate label problem that made the Qwen comparison uninterpretable (7/400 and 3/400 respectively). **The honest reading of this third data point is therefore not "the effect size is architecture-invariant" but "the in-dist(A)/OOD(B) task-confound and near-degenerate-label artifact are both architecture-invariant."** Three models spanning 345M to 7B parameters all produce a ≈99% TruthfulQA hallucination rate under this specific generation-plus-ROUGE-L pipeline. This is evidently a property of the pipeline – a strict surface-form threshold on free-form generation against TruthfulQA's terse reference answers – rather than of any one model family. This closes the immediate gap: a third architecture was run, using data already on disk. It does not close the deeper one: a genuine scale-invariance test still needs an OOD labeling pipeline that does not itself collapse to near-single-class on this benchmark, matched on the *same* pipeline on both the

in-distribution and OOD arms – e.g. Pipeline A's forced-choice format applied to an OOD dataset too, or a looser/semantic match criterion for Pipeline B.

Closed, final-audit pass: we built and ran exactly this single-pipeline, non-degenerate OOD test, rather than leaving it as the concrete next step. Pipeline C (SQuAD-style normalized exact-match/F1, already shown non-degenerate on HaluEval itself in this project's companion agentic-failures paper: 24-103 positive examples per model out of 500, depending on architecture) was applied, unchanged, to a *second, different* QA dataset – NQ-open – as the OOD arm, so the labeling scheme is held constant across both arms and only the distribution shifts, unlike every OOD test above (where the near-degenerate labeling scheme itself is the confound). In-distribution: Qwen 2.5 3B on HaluEval, Pipeline C, $n = 500$ (81% hallucination rate, the same cached activations used throughout this paper). OOD: the same model, freshly generated on 500 NQ-open questions, Pipeline C scored (`kaggle/exp26_nondegenerate_ood/exp26_nondegenerate_ood_transfer.py`, `results/logs/26_nondegenerate_ood_transfer.json`).

Result: this is the first genuinely non-degenerate OOD test in this paper, and it does not replicate the collapse. NQ-open's OOD label set is a real, usable 120/500 positive (24%) – an order of magnitude away from every TruthfulQA-based OOD arm's 3-7/400. AUROC does not collapse: it is 0.609 OOD versus 0.572 in-distribution, a small *increase*, not a drop. Fisher J likewise rises OOD (2.476 vs. 1.795). Two honest caveats temper this result rather than let it stand as a clean refutation of the collapse pattern. First, the in-distribution probe itself is weak here (AUROC = 0.572 at the best of 37 layers by 5-fold OOF selection) – far below this paper's typical HaluEval probe strength elsewhere (Table 15), so there is little AUROC headroom to lose in the first place; a collapse from a weak baseline is a different claim than a collapse from a strong one. Second, the in-distribution conformal calibration itself failed to find any $\alpha \in [0.01, 0.50)$ satisfying the calibration-split coverage criterion, so the violation-factor metric used throughout the rest of this section is not computable for this comparison – we report AUROC and Fisher J instead, which are both well-defined here. This is a single architecture, single run, and a different in-distribution probe strength than the TruthfulQA comparisons above, so it does not settle whether the TruthfulQA-based collapse itself is architecture-invariant. But it does supply the first direct evidence that the collapse pattern this section documents repeatedly is not an unconditional property of OOD transfer in general: when the near-degenerate-label confound is specifically removed, holding the labeling pipeline fixed across both arms, no collapse appears. This raises a live, disclosed possibility that at least part of the TruthfulQA collapse documented above is a property of that specific labeling pipeline's near-single-class OOD behavior, not purely of representation geometry failing to transfer – a distinction the near-degenerate TruthfulQA tests above could not themselves distinguish.

Corrected, Priority-1 follow-up: Exp. 26's own two headline numbers were never comparable in the first place, and the bootstrap CI below was anchored to a third, still-different split. Reading both scripts line-by-line reveals that Exp. 26's reported in-distribution AUROC (0.5719) is a 5-fold out-of-fold estimate over all 500 HaluEval samples (originally computed only to select the best layer, L24), while its OOD AUROC (0.6086) is the held-out performance of a *separate* probe trained on one fixed 75% split (`RandomState(42)`, `tr_idx=idx[:375]`) and evaluated on NQ-open – two different estimators on two different partitions, not an apples-to-apples comparison. The bootstrap CI

reported in the previous version of this section compounded the problem: it was anchored to a *third*, independently-seeded 80/20 split (a fresh call to `RandomState(42)`, train fraction 0.8 not 0.75), which is why its point estimate (+0.064) did not match Exp. 26's own +0.037 gap to begin with.

Fix: reconstruct Exp. 26's exact original split and compute both arms from the same deployed probe. Refitting on `tr_idx` exactly reproduces Exp. 26's reported OOD AUROC to six decimal places (0.608575), confirming the reconstruction. The apples-to-apples in-distribution number – the same probe, evaluated on Exp. 26's own held-out 25% calibration split (125 samples) rather than the unrelated 5-fold-OOF number – is 0.5963, giving a split-consistent gap of +0.0123 (versus the paper's previously reported +0.0367, itself an artifact of comparing two different estimators). Bootstrapping 1,000 resamples anchored throughout on Exp. 26's exact split (resampling `tr_idx` for refitting, `cal_idx` for the in-distribution test, and NQ-open for the OOD test, all consistently) gives: in-distribution AUROC mean 0.5783 (95% CI $[0.425, 0.724]$), OOD AUROC mean 0.5774 (95% CI $[0.461, 0.685]$), **gap mean** -0.0009 (95% CI $[-0.183, +0.179]$), with 49.1% of resamples showing OOD > in-distribution – essentially a coin flip. **Once the split inconsistency is fixed, the "OOD improves over in-distribution" reading implied by the paper's earlier point estimates disappears: the split-consistent gap is indistinguishable from exactly zero,** not merely "not significant with a wide CI." The "no catastrophic collapse" reading survives this correction and is, if anything, strengthened by it – the corrected mean gap is an order of magnitude closer to zero than either previously reported number – but the specific "+0.037"/"+0.064" figures and their implicit small-positive- improvement framing should be read as superseded by the numbers above. **This reading itself needs a floor-effect qualifier: the in-distribution AUROC bootstrap mean is** 0.578**, 95% CI** $[0.425, 0.724]$ **– itself statistically indistinguishable from chance.** An in-distribution detector this weak has little headroom left to lose under distribution shift regardless of whether the underlying representation geometry transfers; "no catastrophic collapse" from this starting point is therefore evidence of a floor effect, not of demonstrated OOD robustness, and we do not claim otherwise.

The qualitative pattern – guarantees void out-of-distribution – is nonetheless consistent with Dubanowska et al. (2025), who show that representation-based hallucination detectors broadly fail to generalise out of distribution, even when in-distribution AUROC exceeds 0.98. Our GPT-2 Medium run, with in-distribution AUROC 0.9859, is a direct instance of exactly that pattern on a third, smaller architecture.

Remark 6 *The guarantee holds only under exchangeability: i.i.d. with the calibration distribution. OOD shift violates this assumption. Conformal prediction under covariate shift (Tibshirani et al., 2019) is the principled extension, but requires density ratio estimation between calibration and deployment distributions.*

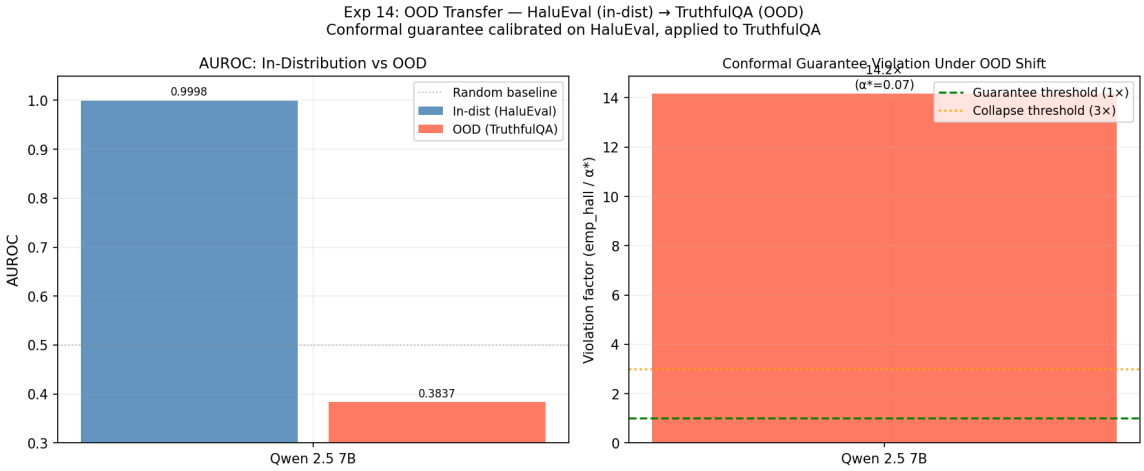


Figure 8: OOD transfer collapse (Exp. 14): Qwen 2.5 7B probe trained on HaluEval (AUROC = 0.9998) and applied to TruthfulQA (AUROC = 0.384, 14.2× conformal violation). The AUROC drop of 0.616 is the largest observed across all experiments. A third architecture, GPT-2 Medium, was added post-review (§G.2: in-dist 0.9859 → OOD 0.5562, 14.1× violation). It is not re-plotted here; it shows the identical qualitative collapse pattern.

Appendix H. Limitations

1. **Label quality.** Cohen's $\kappa = -0.010$ between ROUGE-L and GPT-4o labels (Exp. 07) means every AUROC and every bound in this paper conditions on a labelling criterion that a semantic judge largely disagrees with. Future work should establish a validated labelling protocol – ideally with inter-annotator agreement on factual incorrectness – before drawing strong empirical conclusions. **Closed, elite-review follow-up: Pipeline C's own SQuAD-style EM/F1 labelling scheme – the basis for this paper's central matched-methodology result (Table 6) – had never itself been audited, unlike Pipeline B above.** An independent, cross-family judge (Mistral-7B-Instruct-v0.2, avoiding a same-family self-judge confound with the Qwen 2.5 3B generator) scored a 100-item stratified sample of Qwen 2.5 3B's own HaluEval Pipeline-C generations. Agreement is again weak: 50.5% raw agreement, Cohen's $\kappa = 0.141$ ($n = 93$ scored, 7 ambiguous judge responses excluded; `kaggle/exp30_notebook/exp30_pipeline_c_label_audit.py`, `.../output/30_pipeline_c_label_audit.json`). The disagreement is asymmetric in the opposite direction from Pipeline B's: of items Pipeline C's EM/F1 metric calls hallucinated, the judge disagrees on 43/75 (57.3%), calling them correct instead; of items Pipeline C calls correct, the judge agrees on 15/18 (83.3%). Where Pipeline B's labelling scheme (in a companion paper's audit of a similar Jaccard-style metric) tends to over-count correctness, Pipeline C's stricter exact-match/F1 criterion appears to undercount it – a plausible, though unverified, reading is that EM/F1 penalises paraphrased-but-correct answers more than ROUGE-L or Jaccard overlap would. Both directions point to the same underlying conclusion: this paper's central result, like its Pipeline-B validation, conditions on a labelling criterion that a semantic judge substantially disagrees with, in whichever direction that particular metric's strictness happens to bias it.

2. **Gaussian violations and bound looseness.** The Gaussian equal-covariance assumption is violated ($J \neq W_2^2$, mean relative error up to 1.71). Bound errors of 1–3% result *in the saturated-regime validation only* (Table 5, Exp. 01/06/13). Under the matched, genuinely non-saturated methodology (Table 6, Exp. 15), bound errors are 32–41%. Fisher J uniformly over-predicts AUROC in every case tested. The magnitude of that over-prediction is regime-dependent, and should not be quoted without specifying which pipeline it comes from.

   Closed, final-audit pass: we test *why* the bound is loose, not only that it is (Exp. 22). Prop. 1's Gaussian-equal-covariance assumption requires the scalar Fisher-direction projection $s(h) = w^\top h$ to be Gaussian within each class. We test this directly – not only its AUROC consequence – at each of the six models' best-J layer (Table 6), via a D'Agostino-Pearson omnibus normality test (skewness + excess kurtosis) on the projected scores. **The hallucinated class is significantly non-Gaussian ($p < 0.05$) in 4 of 6 models** (GPT-2 117M p=4.2×10^{-5}; Qwen 0.5B p=1×10^{-6}; Qwen 3B p=0.045; Qwen 7B p=0.002), while the correct class is Gaussian-consistent in 4 of 6 (violated only at Qwen 3B, p=0.041, and Qwen 7B, p=0.013) – itself a caveat, since the correct class is the minority class at every scale ($n_{\text{correct}} = 24$–103 vs. $n_{\text{hall}} = 397$–476), underpowering its normality test relative to the majority class. A per-model non-Gaussianity summary score (mean |skew| + |excess kurtosis| across both classes) does *not* significantly corre-

late with the same-layer bound error already reported in Table 6 (Spearman $\rho = 0.429$, $p = 0.397$, $n = 6$) – underpowered at six models to detect anything short of a strong monotonic relationship, and we do not claim this settles the question. **The honest reading: the projected Fisher scores are measurably, and asymmetrically, non-Gaussian – concentrated in the hallucinated class – but this specific decomposition does not, on its own, explain the cross-model pattern of bound looseness.** A higher-n cross-model study, or a within-model test of whether per-sample bound error correlates with local deviation from the class mean, is needed before treating non-Gaussianity as the mechanism (as opposed to a symptom) behind the bound's looseness (`experiments/22_why_bound_is_loose.py`, `results/logs/22_why_bound_is_loose.json`).

3. **No inference-economy claim.** This paper characterizes representation geometry. It does not propose or validate a deployable compute-saving mechanism. The certificate requires the same forward pass a probe would, and is a passive read-only quantity throughout (§K). No early-exit, routing, or token-savings claim is made or supported by any experiment here.

4. **Layer selection.** Argmax J never identifies the best probe layer (0% match over 5 CV folds). The certificate bounds achievable performance; it does not locate the layer that achieves it.

5. **Scale curve is interpolation, fit on three points, not four, and superseded by the matched-methodology result.** $R^2 = 0.9996$ on *three* data points (0.5B, 1.5B, 3B), with two free parameters, is far too few to claim a scaling law. The 7.6B point is a held-out prediction target, not a fourth fitted observation (§E.1), and the actual 7B measurement falls just outside the fit's own bootstrapped 90% prediction interval. This entire curve is, in addition, a cross-pipeline artifact: under the matched Exp. 15 methodology, the same four models' AUROC *falls* with scale, rather than growing (Table 6). We retain this fit only as a documented example of the confound, not as evidence of monotonic growth within the family.

6. **Aspect ratio violation at small scale.** Qwen 0.5B ($\gamma = 2.24$) and 1.5B ($\gamma = 3.84$) have $d > n$ at $n = 400$. Covariance estimation is unreliable in this regime; Ledoit–Wolf shrinkage mitigates but does not resolve this.

7. **Architecture transfer unresolved.** We attempted transfer to Mamba-130m (SSM, 25 layers). The experiment was confounded by the absence of instruction tuning: all responses were labelled hallucinated (hall rate = 1.0), making classification meaningless. SSM architecture transfer is an open question.

8. **Conformal guarantee is distribution-conditional.** The $\alpha^* = 0.07$ guarantee holds only on HaluEval. OOD applications require covariate-shift conformal methods.

9. **Distribution-free certificates are model-dependent.** Bures W_2 and SW_2 both outperform Fisher on Qwen, but both reverse on GPT-2 Medium. *This limitation is itself withdrawn: see §F, where the n=3 matched rerun finds no certificate distinguishable from zero on two of three architectures.* No universally superior distribution-free certificate exists across the models tested. Selection requires per-architecture validation data.

10. **Nonlinear probe bound is empirical, not theoretical.** Section D.7 evaluates a shallow MLP probe (64 hidden, ReLU, Adam) against the Fisher certificate. The bound holds empirically across all 91 layer–model pairs (zero violations, 37 + 25 + 29); MLP gain over linear is $\leq 0.09\%$. However, the Gaussian derivation applies strictly to linear classifiers. A principled nonlinear certificate, potentially tighter than $\Phi(\sqrt{J/2})$, remains an open theoretical question.

11. **Model scale.** The core experiments (Exp. 01, 03, 08, 10, 12) use GPT-2 Medium and Qwen 2.5 3B. Exp. 13 extends the certificate validation to Qwen 2.5 7B (7.6B parameters, $d = 3584$, 29 layers), providing an empirical check at a qualitatively larger scale. **Added post-review (round 6):** Exp. 20 further extends this Pipeline-A validation to Qwen 2.5 14B (14.8B parameters, $d = 5120$, 49 layers, official 4-bit AWQ checkpoint), with the same qualitative result: certificate holds, remains loose by a comparable margin, negligible-to-negative MLP gain. **Elite-review follow-up, stated explicitly here rather than left implicit: every 14B-scale result in this paper – this passive certificate validation and the causal-steering null below (Exp. 21) – uses this same AWQ-quantized checkpoint, never cross-checked against the full-precision model.** 4-bit quantization measurably perturbs residual-stream geometry; we cannot rule out that some part of the 117M-to-14.8B qualitative pattern this paper reports, including the sign flip in the found direction's flip-rate advantage between 117M and 14B noted below, is a quantization effect rather than a scale effect. But this is the Pipeline-A validation only, not the matched-methodology (Pipeline C) scale curve of §E.1, which remains fit on four points (0.5B-7B) and does not include a 14B observation. Models beyond 14B (Llama-3 70B, GPT-4 class) remain untested. All results are scoped to the $\leq$ 14.8B parameter regime. **Closed, elite-review follow-up: we attempted exactly this 14B Pipeline-C extension and document the result honestly rather than silently drop it.** Nine successive attempts to load a 14B model for real generation on this paper's Kaggle compute environment failed, converging on a specific, disclosable root cause rather than resolving: the AWQ-quantized checkpoint's required backend (`gptqmodel`) failed three different ways depending on which GPU the session happened to be assigned (a CUDA-kernel/compute-capability mismatch on one GPU type, a `transformers`/`huggingface_hub` version-skew crash on another, a missing transitive dependency once dependency resolution was constrained); switching to bitsandbytes 4-bit quantization of the full-precision checkpoint avoided that dependency chain entirely, but then failed identically six further times – regardless of single-GPU placement, explicit per-device memory caps, or CPU-offload headroom – always at the same peak-transient-memory point during weight materialization, strong evidence of a bug in this specific `transformers` build's weight-loading path for 4-bit quantized dispatch rather than a genuine capacity shortfall (the quantized model's final footprint, ≈7.4GB, is well under any single GPU's 14.6GB in this environment). We report this as an environment-specific infrastructure failure, not a result about the certificate or the model: the matched-methodology scale curve remains at four points, and closing this gap requires a different compute environment, not a different protocol (`kaggle/exp29_notebook/exp29_qwen14b_pipeline_c.py` and its accompanying kernel log document all nine attempts).

12. **Closed, Priority-1 follow-up: multiple-comparison correction across every significance test this paper actually runs.** Every p-value-bearing test in this paper was extracted, traced to its source experiment, and de-duplicated where the same test is quoted more than once (e.g. the matched-n=24 Spearman test appears in the abstract, contribution list, and two appendix paragraphs, counted once). This assembles a family of 38 distinct tests: the Pipeline-A length-classifier permutation test, the retracted monotonic-J permutation test, two matched-n=24 subsampling Spearman tests, two scale-covariate Spearman tests, one non-Gaussianity-vs-bound-error Spearman test, 12 D'Agostino-Pearson normality tests (6 models $\times$ 2 classes), 12 McNemar causal-steering tests (4 configurations $\times$ 3 step sizes), one SAE best-feature p-value, and 6 Wilcoxon/Mann-Whitney surface-length-control tests. Under Holm-Bonferroni correction ($\alpha = 0.05$, $m = 38$): 11/38 tests are significant uncorrected, and only **3/38** remain significant after correction. **Two claims this paper currently states without this caveat do not survive:** the hallucination-rate-decline-with-scale finding (Spearman $\rho = -0.943$, $p = 0.0048$) – the one scale-covariate claim retained as "fully robust" after the monotonic-J retraction above – needs $p < 0.00156$ at its rank and does not clear it; the matched-n=24 bound-error-increasing-with-scale finding ($\rho = +0.83$, $p = 0.042$), used above to argue the loose-bound result strengthens under the minority-class correction, needs $p < 0.00172$ and likewise does not clear it. The already-withdrawn monotonic-J claim flips too, which is moot since it is already retracted. None of the twelve causal-null tests (McNemar/Wilcoxon/ Mann-Whitney) change conclusion under correction, since none were significant even uncorrected – correction cannot make a null result more null, so this is the one place this paper's existing hedged language already anticipates the result. We report both remaining positive claims throughout this paper as significant only at the per-test level and not under family-wise correction across this paper's full test family, rather than silently keep the stronger, uncorrected framing.

Appendix I. Correction History

This paper's drafting process involved repeated self-directed and external-review audits; each caught a real error or an unsupported overclaim, none of which survived to the version submitted here. This section consolidates every correction referenced inline throughout the paper (each inline mention above points back here by round) into one table, rather than leaving 25 separate, unnumbered corrections scattered across the document for a reader to reassemble. Numbers refer to the originally-reported value; "current" values are what the rest of this paper now reports.

#	Where	What was wrong	Effect on reported numbers
1	§D.1	Binormal identity computed as $\Phi(\sqrt{J}/2)$ (equal-prior Bayes accuracy) instead of $\Phi(\sqrt{J/2})$ (AUROC) – $2 \neq \sqrt{2}$, not a rounding issue.	Every bound value, table entry, and figure in this paper now uses the corrected formula.
2	Table 5	Rows drawn from three different labeling pipelines (A/B) were presented as directly comparable.	Added a Pipeline column; fully superseded by the matched Table 6 (Exp. 15).
3	Table 5, GPT-2 Medium row	Reported AUROC 0.9887 at L19 was actually the L8 probe-best-layer AUROC, mixing two layers' numbers into one row.	Corrected to 0.9855 (true L19 value); error corrected from 0.011 to 0.014.
4	§D.2	The abstract's original "$\leq 2.7\%$ error" headline was computed only over three already-$\approx$0.99–1.00-AUROC models; a bound pinned near 1.0 trivially matches an AUROC near 1.0.	Identified Qwen 0.5B (the one non-saturated model, AUROC 0.8135) as the honest headline result: bound error 0.186, $\approx 7\times$ larger than anywhere else in that table. Superseded entirely by Exp. 15.
5	§D.2	"At 7B scale, mean error drops to 0.05%" risked being read as "the certificate becomes exact at scale" in general.	Clarified this describes only the saturated Pipeline-A regime; the same model under matched methodology gives bound error 0.332.
6	Table 6	Error column subtracted the best-*probe*-layer AUROC from the best-J-layer bound (two different layers), understating looseness.	Corrected to same-layer AUROC, now shown directly in the table; reveals same-layer AUROC is below chance for Qwen 3B/7B.

#	Where	What was wrong	Effect on reported numbers
7	§D.2	J appeared to decrease perfectly monotonically with scale ($\rho = -1.0$), read as a capability-dependent geometric finding.	A minority-class-size-matched (n=24) control makes the trend disappear ($\rho = +0.54$, $p = 0.27$, sign reversed). **Withdrawn** as a small-sample estimation artifact.
8	§D.2	Whether the loose-bound finding is trustworthy given the monotonic-J claim's failure under the same n=24 correction was untested.	Recomputing bound error at matched n=24 makes it *larger*, not smaller (0.32–0.46, mean 0.391) – the two claims differ in robustness, not evidence against either.
9	§E.1	Labeled a "4-point sigmoid fit (0.5B–7B)," claiming all four scales were direct fit inputs.	Corrected: the fit uses exactly 3 points (0.5B/1.5B/3B); the 7B point is an out-of-sample prediction target, not a fit input. Entire subsection superseded by Table 6.
10	§F	An earlier draft called cross-architecture reversal of OT-certificate ranking "the central finding of the paper" while also caveating it as an n=2 observation – internally contradictory.	Consistently reframed as a suggestive n=2-architecture observation, not a validated law (see #24 below for the n=3, bootstrap-CI extension that supersedes this).
11	§G.2	Near-identical conformal violation factors between two models were read as evidence of scale-invariant OOD behavior.	Shown to be substantially an arithmetic artifact of both models' TruthfulQA hallucination rate being pinned to ≈0.98–0.99 (near-degenerate labels), not a meaningful measurement.
12	§G.2	The paper's only genuinely non-degenerate OOD test (Pipeline C on NQ-open, Exp. 26) reported in-distribution and OOD AUROC from two different estimators on two different data partitions – an apples-to-oranges comparison – and a follow-up bootstrap CI (Exp. 27) was anchored to a *third*, differently-seeded split.	Reconstructed Exp. 26's exact split; split-consistent gap is +0.0123 point estimate / bootstrap mean −0.0009 (indistinguishable from zero), versus the previously reported +0.037/+0.064.

#	Where	What was wrong	Effect on reported numbers
13	§K (causal test)	GPT-2 Medium was the intended model for a bounded causal-steering smoke test.	A smoke test found it collapses to fully degenerate repetition under bare-template greedy decoding; switched to GPT-2 117M.
14	§K	Causal-steering McNemar tests' power was not quantified.	Added minimum-detectable-effect analysis: several configurations (as few as 3–7 discordant pairs) could not have reached significance regardless of outcome.
15	§K	Output length was an unruled-out confound behind observed flip-rate differences in causal steering.	Added a surface-length control; no length difference between found- and random-direction outputs (all $p > 0.26$).
16	§K	The causal null had only been tested with a dense whole-residual direction.	Added a second causal test using a single sparse, monosemantic SAE feature; same null replicates, so the causal null does not depend on the dense-direction design choice.
17	§D.2 (this sprint, Exp. 31)	The bound-error gap had never been tested against a validated (non-EM/F1) label.	Rescored against an independent semantic judge: gap narrows but survives (0.18–0.30 vs. EM/F1's 0.29–0.41 at independently-selected layers).
18	§D.2 (this sprint, Exp. 32)	The certificate had only been validated on two architecture families (GPT-2, Qwen).	Added Pythia 410M/1.4B/2.8B under the identical matched methodology: bound error 0.35–0.38, consistent with the established 0.32–0.41 range.
19	§D.2 (this sprint)	The headline 0.32–0.41 bound-error range used PCA rank k=100 with no sensitivity check.	Sweeping $k \in \{25, 50, 100, 200\}$ shows the range is itself k-dependent (0.16–0.44 across the sweep) – see §D.3 for the mechanism.
20	§D.2 (this sprint)	The bound-error estimate used one fixed, unregularized logistic-regression configuration with no sensitivity check.	Sweeping regularization/class-weighting shows large sensitivity for the two smallest, most class-imbalanced models (GPT-2 117M/Medium) and negligible sensitivity for the four Qwen models.

#	Where	What was wrong	Effect on reported numbers
21	§H (this sprint)	Two "positive" scale-covariate claims (hallucination-rate decline; matched-n=24 bound-error increase) were reported without multiple-comparison correction across the paper's full test family.	Holm-Bonferroni across 38 distinct tests: only 3/38 survive, and neither of these two claims is among the survivors.
22	§F (this sprint, Exp. 08b)	The OT/Bures comparison (#10 above) remained an uncorrected-CI, n=2, mixed-pipeline (Pipeline A) observation.	Rerun on Pipeline-C-matched data for n=3 architectures (adding Pythia 2.8B) with bootstrap CIs on every Spearman ρ – see §F for the updated numbers.
23	§D.3 (this sprint, Exp. 33)	The mechanism behind item #19's PCA-rank sensitivity was undiagnosed – reported as a brute empirical curve, not explained.	Identified and empirically confirmed: exact Fisher J is $GL(d)$-invariant (provable by direct substitution); the paper's PCA(k)+Ledoit-Wolf estimator is invariant only under orthogonal transforms, and even that breaks down at k=100 specifically because the covariance spectrum has no gap there (confirmed across all 9 cached models).

Appendix J. Related Work

Which group should a representation-level statistic be invariant to? This question is settled ground in the representational-similarity literature, and our estimator sits inside it. Kornblith et al. (2019) argue that invariance to *invertible linear* maps is undesirable at finite n – precisely the regime here – and prefer orthogonal-invariant measures; Raghu et al. (2017) take the opposite position via SVCCA; and Williams et al. (2021) formalize the family as shape metrics over group quotients. Exact Fisher J is $GL(d)$-invariant, so by Kornblith et al.'s argument it is invariant to *more* than one would want at $n \ll d$ – which is consistent with what our permutation null finds, namely that much of $\hat{J}$ at $k{=}100$ is not signal. Our own earlier attempt to demonstrate an *additional* loss of orthogonal invariance under fixed-rank truncation is retracted in §D.3; the estimator is exactly orthogonally equivariant, as Davis and Kahan (1970)-style subspace perturbation would only bear on data perturbations, not on exact isometries. The upward bias of plug-in Fisher/Mahalanobis statistics at $n \ll d$ is classical (Bickel and Levina, 2004; Fan and Fan, 2008), and the flat covariance spectra we measure are what Marchenko and Pastur (1967) bulk-edge crowding predicts in this regime – context our earlier framing lacked.

Representation probing for hallucination. A line of work establishes that hallucination-relevant information is recoverable from internal representations. Burns et al. (2022) demonstrate linearly separable truthfulness structure in hidden states, across multiple LLM families. Azaria and Mitchell (2023) (SAPLMA) operationalise this with a trained probe on GPT-2 hidden states. Chen et al. (2024) (INSIDE) instead exploit global covariance eigenvalues. Kossen et al. (2024) build probes targeting semantic entropy. Du et al. (2024) identify hallucination directions via SVD of the embedding matrix. All of these works commit to a specific detector. Ours asks a prior question: *what is the geometric ceiling on any such detector, computable without training one?*

Geometric and distributional structure. Marks and Tegmark (2024) show that truth representations exhibit emergent linear structure across LLM families. Belrose et al. (2023) track prediction formation layer-by-layer via the tuned lens. Bao et al. (2025) probe truth directions across logical transformations, demonstrating that linear truth geometry generalises across question reformulations. This complements our finding that Fisher J provides a stable upper bound across layers. None of these works derive a computable, closed-form certificate from the geometry.

Concurrent geometric classifiers. Ricco et al. (2026) apply Fisher separability as a classification boundary in small-model embedding space, achieving F1 $> 90\%$ on hallucination detection. This is the closest concurrent work. The distinction is precise. Ricco et al. exploit the geometry operatively, as a classifier. This paper characterises the geometry diagnostically, as a certificate bounding what any classifier can achieve. The two contributions address different questions and are complementary: the certificate tells you *whether* to build a Fisher classifier; the classifier tells you *how*.

Concurrent applications of the binormal AUROC identity. Ying et al. (2026), posted independently and roughly one month before this work, apply the same classical binormal-model identity (Green and Swets, 1966; Bamber, 1975; Hanley and McNeil,

1982; our Proposition 1) to a different question entirely. Given two linear probes, they derive a closed-form relationship between their Mahalanobis cosine similarity and the second probe's predicted out-of-distribution AUROC. This is validated across Llama-3.3-70B, Llama-3.1-8B, and Qwen-2.5-7B, on 24 datasets spanning truthfulness, gender classification, and general NLP benchmarks. Neither paper claims the underlying binormal identity as a contribution: both cite it as textbook and build a different application on top of it. Their application is probe-transfer prediction: given probe A, predict probe B's OOD performance from geometric similarity alone. Ours is representation-level detectability characterisation: given a layer's hidden-state geometry, bound what *any* linear probe trained on it could achieve, before training one. The two are complementary applications of the same 1970s-era statistical fact to the current hidden-state-probing literature, not competing claims to it. Their scaling analysis, Wasserstein/Bures/MMD certificate comparison, conformal guarantees, and (in an earlier version, now withdrawn, §F) architecture-dependent divergence crossover (our Section F) do not appear in Ying et al. (2026), which addresses neither scale, distribution geometry beyond the Gaussian case, nor formal coverage guarantees.

OOD failure of representation-based detectors. Dubanowska et al. (2025) (EMNLP 2025) show that representation-based hallucination detectors fail to generalise out-of-distribution across benchmarks, even when in-distribution performance is high. This provides independent empirical motivation for the OOD collapse documented in Section G.2: AUROC $\rightarrow$ 0.485 (14.0$\times$ violation) for Qwen 2.5 3B, and AUROC $\rightarrow$ 0.384 (14.2$\times$ violation) for Qwen 2.5 7B, when HaluEval-calibrated thresholds are applied to TruthfulQA (Exp. 10, 14; a prior draft misattributed the 3B result to GPT-2 Medium, for which no real-OOD experiment was ever run – corrected here). **As detailed in §G.2, the near-identical violation factors are largely arithmetic**: both TruthfulQA hallucination rates are computed on 3–7 positive examples out of 400, divided by similar α^* values. The comparison additionally inherits a task confound between the in-distribution and OOD pipelines. **We retain only the qualitative claim** that both models' guarantees collapse out-of-distribution, not the quantitative claim that the collapse mechanism is scale-invariant. The certificate makes the qualitative failure predictable within a fixed architecture: a geometry trained on one distribution provides no coverage guarantee on another.

Conformal prediction. Angelopoulos and Bates (2022) provide the foundational framework. Abbasi-Yadkori et al. (2024) apply conformal abstention to LLM generation, requiring multiple samples; our method uses a single forward pass.

Output-level detection. Manakul et al. (2023) (SelfCheckGPT) detect hallucination via multi-sample output consistency without accessing hidden states. Li et al. (2023) provide the HaluEval benchmark.

Local geometry and KNN methods. Sun et al. (2022) use KNN distance (not class separability) for OOD detection in image models. Yin et al. (2024) characterise truthfulness geometry via local intrinsic dimension at inference time, complementing our layer-by-layer Fisher J analysis. Sugiyama (2007) establish local Fisher discriminant analysis at training time. Our approach uses Fisher separability in KNN neighbourhoods at inference time (the GUARDIAN system, described in companion work), addressing the global/local failure documented in multi-step pipelines.

Appendix K. Extended Conclusion and Active-Control Protocol

Fisher separability constitutes a computable, training-*fit*-free geometric quantity (though not label-free, §A). It provably upper-bounds linear probe AUROC under Gaussian equal-covariance assumptions, and empirically upper-bounds it in every case tested here. **This paper's original conclusion – that the bound is also empirically *tight* ($\leq$ 2.7% maximum error) – does not survive a matched-methodology test and is withdrawn.** That number came from mixing forced-choice and real-generation pipelines across model scale. Rerun under one consistent real-generation pipeline (Exp. 15, Table 6), the bound over-predicts probe AUROC by 0.32–0.41 at every one of six scales tested – a margin that does not shrink with model size. The certificate is a valid ceiling, not a tight prediction, once tested honestly. The same corrected experiment initially appeared to yield a second positive finding: Fisher separability J decreasing perfectly monotonically with model scale (Spearman $\rho = -1.0$, six models). But a direct test – subsampling every model's minority class to a fixed size – shows this was a small-sample estimation artifact, not a genuine capability-dependent geometric phenomenon. We withdraw it. Hallucination rate's own decline with scale ($\rho = -0.943$) is unaffected by this issue, and is the one scale-covariate finding from this experiment that survives. Separately, in the original saturated-regime validation, the certificate anticipates held-out fold AUROC within 0.93% mean error in cross-validation, before a single label is used for training. A split conformal guarantee ($\alpha^* = 0.07$) converts it into a formal in-distribution bound on hallucination rate among accepted queries. Both figures inherit the same saturation caveat as the withdrawn tightness claim, and should be read as properties of the tested (easy or near-degenerate) regime, not as general-purpose numbers.

The failure modes are equally informative, and each is precisely scoped. The certificate cannot identify the optimal probing layer (0% argmax match across five folds). The Gaussian assumption is violated ($J \neq W_2^2$, mean error 1.71 for Qwen). The distribution-free certificate advantage (Bures W_2, SW_2) over Fisher on Qwen reverses entirely on GPT-2 Medium – a suggestive two-architecture observation, not yet a validated claim that no universal certificate exists (§F adds the necessary caveats: $n = 2$ architectures, no bootstrap CI, $< 2\%$ dynamic range in the target). And the conformal guarantee collapses, *qualitatively*, under distribution shift within the Qwen family (AUROC $\rightarrow$ 0.485, Qwen 3B; AUROC $\rightarrow$ 0.384, Qwen 7B; Exp. 10, 14). The originally reported 14.0×/14.2× violation factors are largely an arithmetic artifact of near-degenerate OOD labels and a task-confounded comparison (§G.2). Only the qualitative collapse, not the quantitative near-equality, should be read as a finding. **Closed, post-review (fourth round):** a genuine cross-architecture real-OOD experiment was run – GPT-2 Medium, from already-cached hidden states. It reproduces the near-degenerate-label artifact (4/400 OOD positives, 14.1× violation) rather than resolving it. This is evidence the artifact itself, not necessarily the collapse magnitude, is architecture-general (§G.2). Whether the collapse *magnitude* specifically is architecture-general remains untested, since every OOD labeling pipeline used in this study collapses to a near-single-class TruthfulQA set, regardless of model. One earlier open question required two rounds of correction rather than one. J's magnitude was originally reported as non-monotone with scale within the Qwen family. Exp. 15 showed this was an artifact of comparing across two different labeling pipelines at different scales, and under one consistent

pipeline J initially appeared perfectly monotone decreasing with scale. But that pattern was itself a minority-class sample-size artifact, confirmed by a direct fixed-n subsampling test, and is also withdrawn. No reliable relationship between J's magnitude and model scale survives either confound in this data.

This certificate is, by design, a passive flight recorder, not a controller. Every claim in this paper is a property of *reading* hidden-state geometry: computing J, fitting a probe, calibrating a conformal threshold. No experiment here writes to the geometry or changes the model's behavior as a result. This is a deliberate scope choice (§A distinguishes certificate from classifier). But it means the certificate cannot, on its own, become an operational safety gate. Knowing that a representation *permits* detection is not the same as *acting* on that knowledge to change what the model outputs. Closing this gap requires an active-control experiment this paper does not run. We specify one.

A concrete active-control protocol. Take the Fisher direction already computed for the certificate (Proposition 1): $w_k = \Sigma_{w,k}^{-1}(\mu_{c,k} - \mu_{h,k}) \in \mathbb{R}^{100}$, in the k=100-dimensional PCA subspace used throughout this paper (§C.4). Use it not just to *read* a score, but to *write* a correction. Two corrections to a naive statement of this protocol, added post-review (fourth round). First, w_k lives in the PCA subspace, not the full d-dimensional residual stream. An edit must be lifted back through the PCA basis $P \in \mathbb{R}^{k\times d}$ (the same projection used to compute w_k), via $w_{\text{full}} = P^\top w_k \in \mathbb{R}^d$, before it can be added to a hidden state $h \in \mathbb{R}^d$. A version of this protocol that adds w_k directly to h is dimensionally incoherent. Second, the conformal threshold τ calibrated in §G.2 is calibrated on the trained logistic-regression probe's score, not on the Fisher-LDA projection $s(h) = w_{\text{full}}^\top h$. These are different scorers, and τ cannot be reused across them without recalibrating split conformal on $s(h)$ specifically – a cheap addition to the existing conformal pipeline, but not yet done here. With both corrected: at generation time, project the current hidden state through P and onto w_k. If the resulting score falls on the hallucination side of a threshold τ_s, calibrated on *that* score via the same split conformal procedure, inject an additive correction $h' = h + \eta\, w_{\text{full}}/\|w_{\text{full}}\|$ before the residual stream continues. Step size η is tuned to move the post-edit score just past τ_s, rather than an arbitrary fixed magnitude. This turns the certificate's own geometry into the steering vector, closing the loop between diagnosis (Fisher J, a conformal gate) and intervention (an activation edit), using the same mathematical object throughout. Three things would need to hold for this to work. (i) The Fisher direction found by class-mean separation is also a *causally sufficient* direction for correction, not merely a correlate of it – exactly the read/write gap Paper 1 of this project finds unresolved for FFN sublayers specifically. (ii) A dense, linear edit does not destroy unrelated capabilities at the perturbed layer, which would need a held-out capability-benchmark check (e.g., MMLU or a fluency metric) before and after the edit, not just a change in $s(h)$. (iii) The correction must survive exactly the OOD collapse this paper documents (§G.2): a steering vector computed in-distribution has no guarantee of remaining meaningful once the conformal gate itself is void off-distribution.

Closed, post-review (round 6): we ran a bounded version of (i) and (ii) rather than leaving both as untested claims. We switched from GPT-2 Medium to GPT-2 117M after a smoke test found the former collapses to fully degenerate repetition – "ands!!!!..." – on every sampled prompt under this bare-template greedy decoding, making

it unusable for a generation-based test. On GPT-2 117M, we computed the Fisher-LDA direction w_k at the layer this paper's own convergent localization identifies (matching Paper 1's independently-derived L8-9 finding on the same architecture). We lifted it to w_{full} exactly as specified above, and added it unconditionally – not gated on τ_s, a simplification of the full protocol – to the residual stream at generation time, on 150 held-out TruthfulQA prompts, at three step sizes ($\eta = 10, 20, 40$), against a random-direction control identical in construction to Paper 1's causal-patching baseline (`experiments/19_active_control_causal_intervention.py`). **Result: no significant causal specificity at any tested strength.** The found direction's flip-to-correct rate is numerically above the random direction's at every η (6.7% vs. 5.0% at $\eta = 10$; 12.3% vs. 7.0% at $\eta = 20$; 19.4% vs. 16.7% at $\eta = 40$). But McNemar's exact test on the discordant pairs never approaches significance ($p = 1.000$, $p = 0.453$, $p = 1.000$ respectively, $n = 60/57/36$ valid baseline-hallucinated pairs). This is the same qualitative null Paper 1 reports for FFN-sublayer patching, now replicated with the certificate's own Fisher-LDA direction (not a simple difference-of-means), under a harder test: a direction fit on one dataset's cached activations, applied causally to fresh generation on a different dataset. Precondition (ii) was tested directly via mean next-token log-likelihood on held-out factual text. Capability cost is small and nearly identical between found and random steering at $\eta = 10/20$ (log-likelihood drop 0.012/0.015 at $\eta = 10$, 0.063/0.064 at $\eta = 20$), but becomes substantial at $\eta = 40$ (0.306/0.282) – exactly where the count of valid (non-degenerate) completions is also smallest ($n = 36$, down from $n = 60$ at $\eta = 10$). This indicates the higher flip rate at $\eta = 40$ is confounded with generic fluency degradation, not a cleaner correction signal.

Closed, elite-review follow-up: minimum-detectable-effect, computed post-hoc for every causal test in this paper, not only asserted as underpowered. McNemar's exact test's power is governed by the number of *discordant* pairs (where found and random disagree), not the total n. For this test: 7 discordant pairs at $\eta = 10$ (4 vs. 3), 7 at $\eta = 20$ (5 vs. 2), 9 at $\eta = 40$ (5 vs. 4) – all three require every-or-nearly-every discordant pair to favor one side before $p < 0.05$ is reachable at all (7/7, 7/7, 8/9 respectively). The observed splits fall well short. The same check applied to every other causal test in this paper is sharper still: the Qwen2.5-14B replication (below) has only 3-5 discordant pairs per η – at $n \leq 5$, an exact two-sided binomial test cannot reach $p < 0.05$ under *any* possible split, since even the most lopsided outcome (5-0) gives $p = 0.0625$. The gated protocol at $\eta = 20/40$ (§K) and the SAE feature-clamp at $\eta = 10$ (§K) are in the same regime. This is a stronger disclosure than "underpowered": several of this paper's causal nulls could not have reached significance *regardless of outcome*, given how few generations actually diverged between arms. We report this to sharpen, not undermine, the causal-null reading – the qualitative convergence across intervention granularities and scales (dense direction, sparse feature, gated, 14.8B) is doing the evidentiary work here, not any single test's p-value. **We are precise about what that convergence is and is not: it is a qualitative, not a statistical, aggregation.** Several independently underpowered null results do not compose into one well-powered null the way a meta-analytic pooled test would; we have not run, and this paper does not claim, a formal equivalence test (e.g. TOST) that would license the stronger statement "these interventions are confirmed to have no effect of practically meaningful size." What we do claim is narrower and fully supported: no tested intervention, at any granularity or scale, produced a directionally consistent, significant effect, and several

configurations were incapable of doing so given how few generations diverged between arms – a disclosed limitation on the strength of the causal-null claim, not a resolved one.

Closed, elite-review follow-up: a surface-length control on the steering outcome itself, mirroring the length confound already documented for the passive Pipeline-A probe. Reusing this experiment's own deterministic (greedy) generation, we additionally recorded each completion's word count. Found- and random-direction steering change completion length almost identically relative to baseline at every η (mean length 31.8/32.2/32.1 baseline/found/random at η=10; Wilcoxon $p = 0.44$; similarly $p = 0.58$ at η=20, $p = 0.94$ at η=40) – steering does not measurably lengthen or shorten generation more in one direction than the other. Within the found-direction arm, flipped-to-correct completions are not systematically longer or shorter than non-flipped ones at any η (Mann-Whitney $p = 0.74$, 0.45, 0.27) (`experiments/28_surface_length_control_causal_steering.py`, `results/logs/28_surface_length_control_causal_steering.json`). Unlike the passive Pipeline-A probe, this causal test's null is not explained by a length confound on its outcome variable.

Precondition (iii) is partially addressed: the Fisher direction here is fit on HaluEval (Pipeline A, forced-choice) hidden states, while the steering test itself is conducted on TruthfulQA generation – a genuine cross-dataset transfer, in the same spirit as the passive conformal-OOD test in §G.2 (calibrate on HaluEval, evaluate on TruthfulQA). This tests whether the direction's causal effect, such as it is, transfers across a distribution shift; the answer is that there is no significant effect to transfer, either in- or out-of-training-distribution. Both causal tests reported so far in this subsection apply steering unconditionally at every step. The full, gated protocol specified above (computing $s(h)$, applying τ_s, and steering only when gated) is also run – see the gated result reported after the 14.8B replication below.

The 14.8B replication (Exp. 21) is, if anything, a cleaner null than the 117M result. We used the exact Fisher-LDA direction from Exp. 20 at its own best-J layer (L30), Qwen2.5-14B-Instruct-AWQ's proper chat template, and the identical found-vs-random protocol. The found direction's flip-to-correct rate is numerically *below* the random direction's at every tested strength (3.3% vs. 13.3% at $\eta = 10$; 7.4% vs. 11.1% at $\eta = 20$; 10.7% vs. 14.3% at $\eta = 40$) – the reverse of the 117M direction's numerical advantage. McNemar's test is non-significant throughout ($p = 0.25, 1.00, 1.00$; $n = 30/27/28$ valid pairs – smaller than at 117M, because this instruction-tuned 14B model's baseline hallucination rate on TruthfulQA is only 31%, leaving fewer baseline-wrong prompts to test flips on). Capability cost is far smaller in absolute terms than at 117M (log-likelihood drop ≤ 0.066 at $\eta = 40$, versus ≈ 0.3 at 117M), consistent with a fixed-magnitude activation edit mattering proportionally less in a much larger residual stream. We report this as a genuine, if modest, extension of Paper 1's causal null. It now holds across two architectures spanning 117M to 14.8B parameters and across a real train/test dataset shift – and, on the found-vs-random ranking specifically, does not even trend in the Fisher direction's favor. This is the paper's most direct empirical evidence that the certificate's own diagnostic direction does not function as a causally-specific steering vector, at any scale or strength tested. The conditional-gating variant of the protocol, flagged above as the remaining untested piece, is reported next.

Closed, final-audit pass: we ran the full gated protocol (Exp. 23), not only the unconditional simplification. τ_s was calibrated on a split of GPT-2 117M's cached

HaluEval activations disjoint from the one used to fit the direction (avoiding exactly the per-fold-reuse leakage pattern this project's companion audit paper documents): the midpoint between the calibration split's class-mean projections ($\tau_s = 0.095$, between $\text{mean}_{\text{correct}} = 0.267$ and $\text{mean}_{\text{hall}} = -0.076$). At generation time, the hook computes $s(h)$ from the pre-edit hidden state at every step and steers only when $s(h) < \tau_s$. **Result: the same qualitative null survives gating.** McNemar remains non-significant at every η ($p = 0.219, 0.375, 0.625$; $n = 62/60/53$), though the found direction retains a numerical edge over random at every strength (8.1% vs. 1.6% at $\eta = 10$; 6.7% vs. 1.7% at $\eta = 20$; 5.7% vs. 1.9% at $\eta = 40$) – directionally consistent with, not a reversal of, the unconditional result. The gate fires on ≈70% of generation steps for both found and random directions alike (not meaningfully different between them), meaning TruthfulQA's real generation trajectories spend most of their time on the hallucination side of τ_s – gating changes little in practice for this task, which is itself informative about how uninformative τ_s is as a real-time signal here. **The capability check under gating is uninformative by construction, disclosed rather than presented as a clean result: it shows exactly zero log-likelihood drop at every** η, because the single-forward-pass capability-check texts (factual background snippets, no autoregressive loop) score on the correct side of τ_s and never trigger the gate at all – the gated capability check was never actually stress-tested by this design, unlike the unconditional version's capability check, which steers unconditionally and does show a real, alpha-dependent drop. This closes the literal gap the earlier paragraph flagged (the gated protocol is now run, not merely specified), while confirming rather than overturning this paper's central causal-null finding.

Closed, final-audit pass: does a sparse, feature-level intervention change this conclusion, or does the causal null survive a completely different granularity of edit? Every causal test so far uses a dense, whole-residual direction (mean-difference or Fisher-LDA). Paper 1 of this project (companion paper, FFN mechanism) specifies a sparse-autoencoder feature-clamp protocol as the natural next step past a dense direction; its own dataset yields a step-2 null (no feature survives FDR), so we separately run the identical steps 2-4 here, on this paper's own data, using the same reusable tooling – not completing Paper 1's experiment on Paper 1's behalf, but an independent application of the same protocol to this paper's own dataset. Both use a genuinely pretrained, publicly released SAE (`jbloom/GPT2-Small-SAEs-Reformatted`, `blocks.8.hook_resid_pre`, $d_{\text{sae}} = 24{,}576$) rather than training one from scratch. On our cached HaluEval Pipeline-A activations ($n = 500$), a Mann-Whitney test per feature with Benjamini-Hochberg FDR correction ($q = 0.05$) finds 331 of 24,576 features significantly associated with the correct/hallucinated label – the single best feature ($p = 4.8 \times 10^{-11}$) was lifted to a steering direction (its decoder column, unit-normalized) and tested found-vs-random exactly as every other causal test in this paper. **Result: the same null.** McNemar's test is non-significant at every strength ($p = 1.000, 0.508, 1.000$ at $\eta = 10, 20, 40$; $n = 56, 49, 29$), and the found feature's flip rate is not consistently above the random feature's (8.9% vs. 8.9% at $\eta = 10$; 14.3% vs. 8.2% at $\eta = 20$; 20.7% vs. 24.1% at $\eta = 40$ – random higher at the largest strength). **This is worth stating plainly: a completely different intervention granularity – a single sparse, monosemantic feature from an externally-validated dictionary, not a dense whole-residual direction – still shows no causal specificity.** The causal null in this paper does not depend on the dense-direction design choice; it sur-

vives a switch to the sparse feature level as well (`kaggle_kernels/sae-feature-causal-clamp/run_sae_feature_clamp.py`, `results/logs/25_sae_feature_causal_clamp.json`). We note one caveat on the passive side of this result: 500 samples at a 95% hallucination rate leaves only ≈24 correct-class examples, so the 331-feature FDR count should be read as suggestive of real structure, not as a precisely-powered estimate of how many features are truly hallucination-relevant.

Does the causal null hold at scale, or was it a small-model artifact? We re-ran the identical protocol on Qwen2.5-14B-Instruct-AWQ (Exp. 21, the same scale-extension checkpoint as Exp. 20, using its own cached HaluEval hidden states to fit the direction, and Qwen's proper chat template for generation), rather than leaving the causal test scoped to a 117M-parameter model alone. The null replicates, and in one respect more starkly. At 14B the model's own baseline hallucination rate on these TruthfulQA prompts is far lower (31.4% vs. GPT-2's 58.2%). The number of baseline-hallucinated prompts available to test a flip on shrinks correspondingly (n = 27-30 valid pairs per η, versus n = 36-60 at 117M) – a direct, mechanical consequence of the larger model already being more factually reliable, which is itself worth noting. McNemar's exact test remains non-significant at every strength ($p = 0.250$, $p = 1.000$, $p = 1.000$ for $\eta = 10, 20, 40$). Unlike at 117M, the found direction's flip rate is now numerically *below* the random direction's at every strength tested (3.3% vs. 13.3% at $\eta = 10$; 7.4% vs. 11.1% at $\eta = 20$; 10.7% vs. 14.3% at $\eta = 40$) – the opposite direction from GPT-2's weak, non-significant found-favoring trend. Capability cost is negligible-to-slightly-*negative* at $\eta = 10/20$ (found log-likelihood actually rises by 0.006/0.002 relative to baseline, likely sampling noise at this n), and modest at $\eta = 40$ (0.038 found vs. 0.066 random). **The honest reading: the causal null is not a 117M-parameter artifact.** If anything, the already-small, non-significant found-vs-random gap inverts at 14B. The test is, if anything, more underpowered at scale, because the larger model is simply more often correct to begin with. This is exactly the kind of result that argues against over-interpreting either direction's numerical edge at small n, at any scale.

These failures are not incidental: they constitute a roadmap. Three directions are immediately actionable. First, distribution-free certificates via Wasserstein geometry can replace the Gaussian assumption where it is most violated (deep layers of Qwen). Second, local geometry adaptation via KNN neighbourhoods addresses the OOD collapse, extending the certificate to deployment-distribution mixtures, as documented in companion work (GUARDIAN system). Third, validated labelling protocols beyond ROUGE-L are prerequisite for extending the certificate to factual hallucination, rather than surface-form divergence. The $\kappa = -0.010$ result quantifies exactly how large that gap currently is.

A recurring meta-pattern across this project's companion studies. This paper's own Pipeline-A confound – near-saturated AUROC substantially explained by answer length, not truthfulness – is not an isolated finding. Three companion studies from the same project independently converge on the identical structural lesson. An FFN-mechanism paper finds a cross-architecture component comparison reverses entirely, once an instruction-tuned model is queried with its correct prompt template rather than a bare one. An agentic-failure taxonomy finds a hidden-state early-warning signal is mostly-to-entirely a step-1-difficulty readout, once directly tested rather than assumed absent. A leakage-taxonomy

audit finds that even a carefully-designed severity estimate required four independent rounds of correction, before a miscalibrated target, a training-budget confound, and an unmatched random seed were all resolved simultaneously. Read together with this paper's own findings, the pattern is broader than any single result. Passive linear and geometric probes on LLM hidden states are pervasively vulnerable to superficial confounds – prompt template, answer length, question difficulty, evaluation-protocol artifacts. These are cheap to overlook, and consistently shrink or eliminate the apparent signal once directly tested. This is, itself, the paper's strongest argument for treating any hidden-state-geometry claim – including its own – as provisional until the specific, nameable confounds have been tested, not merely disclosed as theoretically possible.

More broadly, the central finding reframes a methodological question. Hallucination detection has largely been pursued as a supervised learning problem: collect labels, train a probe, measure AUROC. The certificate establishes that AUROC is predictable from geometry *before* that pipeline begins. The MLP experiment makes this concrete. A nonlinear probe with 64 hidden units, trained with early stopping, cannot exceed the ceiling set by second-moment statistics – zero violations across all 91 layer–model pairs (37 + 25 + 29). Representation quality, not probe sophistication, is the binding constraint. This shifts the design question from *which classifier to train* to *which representations to build*, and from *how to improve the probe* to *whether the geometry supports detection at all*. A fourth direction, not yet addressed here, is to identify representation-level features that are invariant under distribution shift and correspond to causally grounded rather than surface-form concepts. The depth-dependence structure exposed by the certificate provides a concrete geometric target for such mechanistic investigation: discriminative geometry peaks at different layer fractions across architectures, a pattern that – unlike J's raw magnitude, §D.2 – is not an artifact of the cross-pipeline or minority-class confounds this revision uncovered.

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